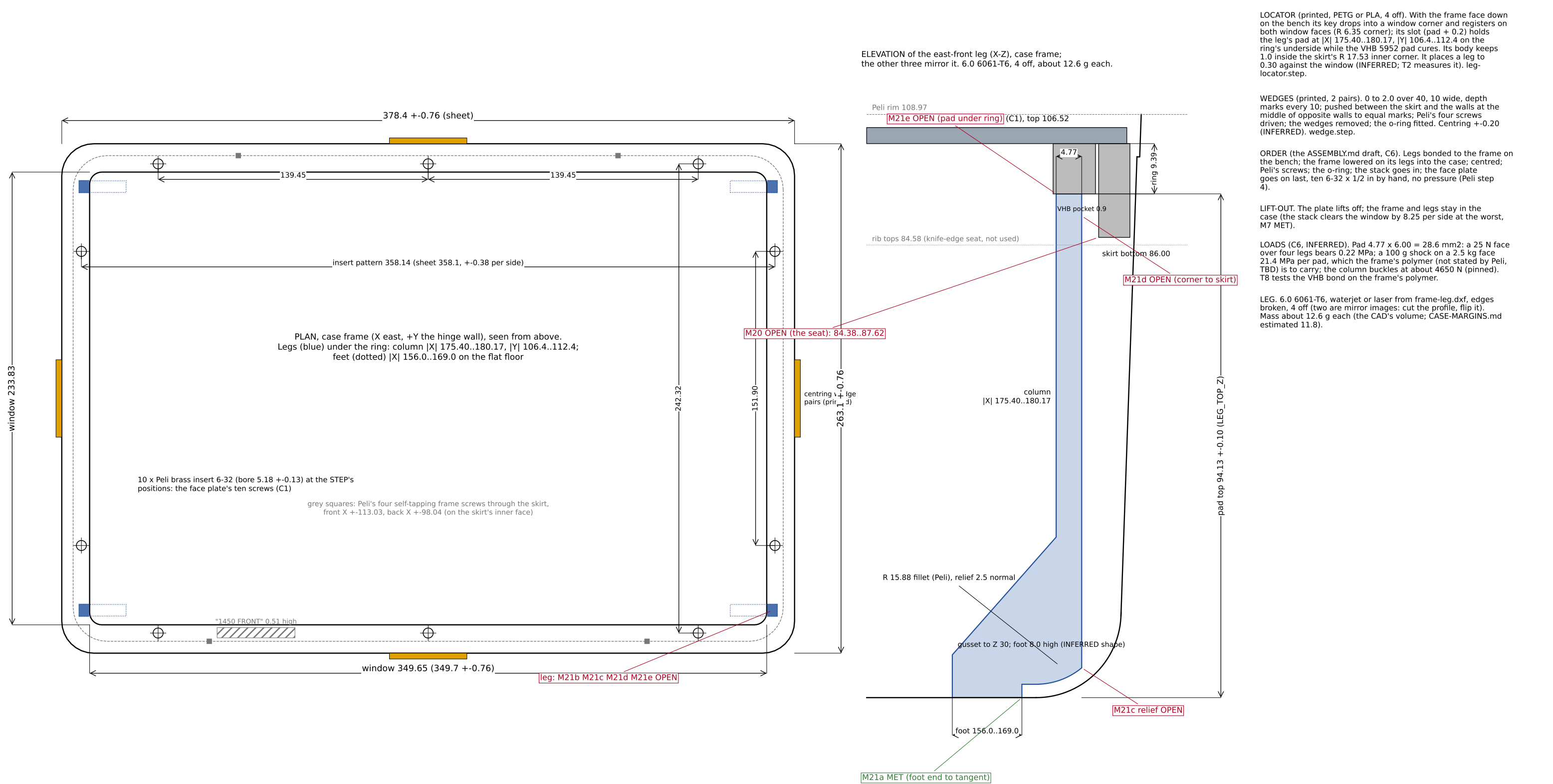




Margin rows this sheet's features enter (frame_seat.out): key, verdict, min, nominal, worst, worst with unstated x2					
M20	OPEN	the frame's seat: bottom Z 86.00 nominal, 84.38..87.62 on the legs (T2)			
M21a	MET	0.00 +2.64 +1.38 +0.50 leg foot on the flat floor: foot end X 169.0 to the fillet tangent			
M21b	OPEN	0.00 +2.09 +0.83 -0.05 leg foot on the flat floor: foot [Y] 112.4 to the long-wall fillet tangent			
M21c	OPEN	1.00 +2.50 +1.24 +0.36 leg relief to Peli's floor fillet (normal)			
M21d	OPEN	1.00 +2.48 +1.14 +0.26 leg column to the frame skirt's inner face: Euclidean, the leg placed in X and Y (outer-back corner)			
M21e	OPEN	0.00 +0.57 +0.27 -0.03 leg pad fully under the ring (pad inner edge to the window edge)			
M21f	MET	1.00 +6.40 +4.52 +3.02 leg to B16's corner (165, 100) in Y			
M21g	MET	1.00 +4.34 +3.08 +2.20 setting leg's column (X 180.17) to the entry plate's screw heads at [Y] 100 (X 184.51), in X			
M21h	MET	1.00 +3.40 +1.51 +1.01 leg column to the backer ring's outer edge (X 172, panel1450.py:93-95)			
M5	OPEN	1.00 +3.65 +1.77 +0.27 pack group (205.5 long, A06) to the east legs' inner faces at [Y] 106.4, per side			
M8z	OPEN	0.00 +2.48 +0.10 -2.18 plate underside above the shoulder, so the plate edge faces the rim zone			
M1	OPEN	2.00 +6.26 +3.62 +1.80 monitor body over the CM5 heatsinks			
M7	MET	1.00 +9.83 +8.25 +7.05 stack lift-out through the frame window, per side in X (Y has 16.91)			

ELEVATION of the east-front leg (X-Z), case frame;
the other three mirror it. 6.0 6061-T6, 4 off, about 12.6 g each.

LOCATOR (printed, PETG or PLA, 4 off). With the frame face down on the bench its key drops into a window corner and registers on both window faces (R 6.35 corner); its slot (pad + 0.2) holds the leg's pad at [X] 175.40..180.17, [Y] 106.4..112.4 on the ring's underside while the VHB 5952 pad cures. Its body keeps 1.0 inside the skirt's R 17.53 inner corner. It places a leg to 0.30 against the window (INFERRED; T2 measures it). leg-locator.step.

WEDGES (printed, 2 pairs). 0 to 2.0 over 40, 10 wide, depth marks every 10; pushed between the skirt and the walls at the middle of opposite walls to equal marks; Peli's four screws driven; the wedges removed; the o-ring fitted. Centring +0.20 (INFERRED). wedge.step.

ORDER (the ASSEMBLY.md draft, C6). Legs bonded to the frame on the bench; the frame lowered on its legs into the case; centred; Peli's screws; the o-ring; the stack goes in; the face plate goes on last, ten 6-32 x 1/2 in by hand, no pressure (Peli step 4).

LIFT-OUT. The plate lifts off; the frame and legs stay in the case (the stack clears the window by 8.25 per side at the worst, M7 MET).

LOADS (C6, INFERRED). Pad 4.77 x 6.00 = 28.6 mm2: a 25 N face over four legs bears 0.22 MPa; a 100 g shock on a 2.5 kg face 21.4 MPa per pad, which the frame's polymer (not stated by Peli, TBD) is to carry; the column buckles at about 4650 N (pinned). T8 tests the VHB bond on the frame's polymer.

LEG. 6.0 6061-T6, waterjet or laser from frame-leg.dxf, edges broken, 4 off (two are mirror images: cut the profile, flip it). Mass about 12.6 g each (the CAD's volume; CASE-MARGINS.md estimated 11.8).

TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines +/-0.10; the 3.0 face plate +/-0.13 and the 6.0 entry plates +/-0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +/-0.10; rebate depth and line +/-0.10; spot-face floors +/-0.10.
Peli: the 1450PF sheet 1453-314-000 rev A states +/-0.76 on one-decimal and +/-0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
The legs' zones ([X] 156.0..180.17, [Y] 106.4..112.4, floor to Z 94.2, each within 0.88 of its place) are theirs alone.

MeshSat field kit V2, Peli 1450 case set

1450PF frame with Peli's insert bores, the four setting legs (C6), locator and wedges

MESH-SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
3bdb88df9826 v2/ecad/tools/panel1450.py
214e915b9855 v2/vendor/peli/1450/frame_seat.out
28c5adc7166 v2/cad/frame_leg.py
f58e434c6105 v2/vendor/peli/1450/1450-panel-frame.STEP

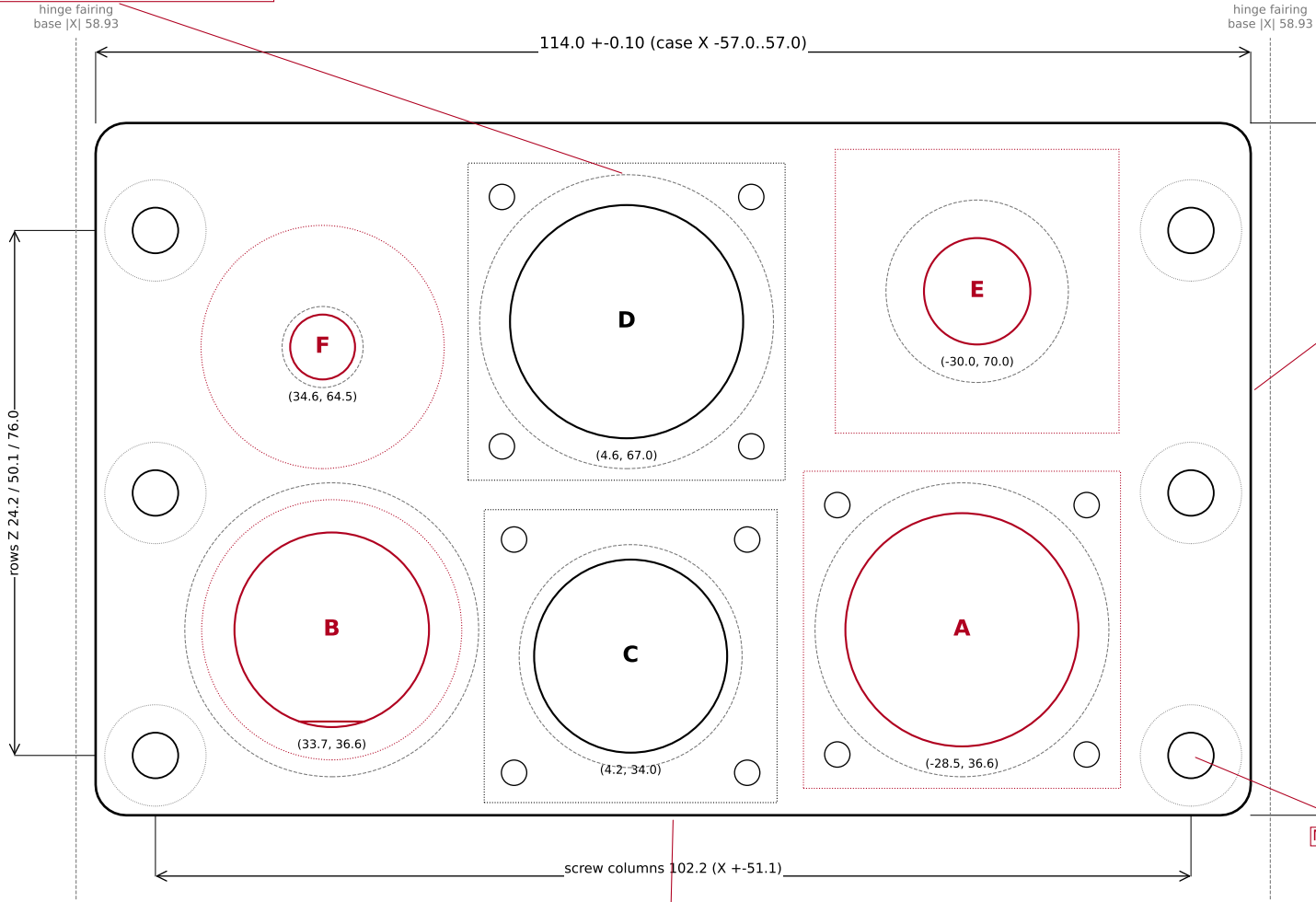
Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 2) from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy.

Sheet 2 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

SEEN FROM OUTSIDE THE BACK WALL: case +X (east) on your LEFT. Solid: plate cut-outs (red: the pick is OPEN, cut-out PROVISIONAL); dashed: the wall hole under it; dotted: the part's flange or body on the plate face; small circles: M3 tapped (drill 2.5).

M14a OPEN (D's body top 81.5 inside, skirt above)



M14i OPEN

M14m OPEN (band)

- ITEMS (panel1450.CONN_ITEMS; centre in case X, Z; plate cut-out / wall hole; flange screws)
- A (-28.5, 36.6) cut-out 23.01 wall 29 4 x M3 on 24.61 sq [OPEN]
sealed RJ45, PoE out: MIL-DTL-38999 shell 15 wall-mount class, 54 V or more (PICK OPEN: the PX0833 fails both)
- C (+4.2, 34.0) cut-out 19.05 wall 22 4 x M3 on 23.01 sq [picked]
shore DC: Glenair D38999/20 shell 13 wall mount, round holes (D0)
- B (+33.7, 36.6) cut-out 19.20 D, flat 9.05 to -Z wall 29 no flange screws [OPEN]
sealed USB-C 45 W outlet, power only: Bulgin 4000 series rear-panel class (PXP4043/C panel drawing OPEN)
- E (-30.0, 70.0) cut-out 10.50 wall 18 no flange screws [OPEN]
outside sensor pod on its M8 receptacle (binder 86 6618 1121 00004 recommended, sheet not held)
- D (+4.6, 67.0) cut-out 23.01 wall 29 4 x M3 on 24.61 sq [picked]
USB host (console, key fill): Glenair 233-370 shell 15 (D0)
- F (+34.6, 64.5) cut-out 6.40 wall 8 no flange screws [OPEN]
ground stud, M6 class: the external earth lead and both RF entry plates' leads outside, the one bonding strap inside

PLATE: 5052-H32 or 6061-T6 aluminium, 5.0, 5.0 thick, R 3.0 corners, edges broken; M4 x 25 A2, a bonded sealing washer 10 across under the head, a plain washer 9.0 and a Nyloc inside.

GASKET: 2.0 closed-cell EPDM or neoprene, the plate's outline, holes = the wall holes (connector-plate-gasket.dxf).

Flanges of A, C and D on Glenair 930-001 flange gaskets (0.76), M3 x 10 A2 into the plate's threads; nothing stands proud of the back.

WALL (1:1 template, back-wall sheet): hole saws 29 (A, B, D), 22 (C), 18 (E), drill 8 (F) and six 4.5; marked +/-0.3 (INFERRED).

STACK THROUGH THE WALL: flange gasket 0.76 + plate 5.0 + gasket 2.0 (1.5 compressed) + wall 5.34 (DXF A-A).

INSIDE: A's patch plug under B16 and U51 and outboard of A22 (M14e, M14f); B's lead bent down at R 28 (M14g, M14h); C's cores turned down within 12 (M14d); D's right-angle USB-A plug, cable down, within 20.0 (M14c); F's bonding strap to A's ground.

OPEN PICKS (section 6 of CASE-MARGINS.md): the sealed RJ45 (a shell 15 envelope rated 54 V: the Bulgin PX0833 fails both), the PXP4043/C panel drawing, the M8 receptacle's sheet, the pod and the stud. Until each is picked its cut-out stays PROVISIONAL.

Margin rows this sheet's features enter (frame_seat.out): key, verdict, min, nominal, worst, worst with unstated x2

M14a	OPEN	1.00	+4.50	+2.58	+0.76	highest part inside the back wall below the frame skirt (D's body; the top screws' washers reach 80.5)
M14b	OPEN	1.00	+2.40	+1.34	+0.28	connector plate bottom edge above the outer bottom radius
M14i	OPEN	1.00	+1.93	+1.15	+0.47	connector plate edge to the hinge fairings' bases in X
M14j	MET	1.00	+1.81	+1.13	+1.13	tightest pair on the plate face (flanges, pod, stud, screw washers), each at its float: A / D
M14k	MET	1.00	+1.85	+1.11	+1.11	tightest pair of mated connectors (shell 15 plugs 32.51, shell 13 plug 29.4), each at its float: D / F
M14l	MET	2.00	+4.11	+3.51	+2.91	tightest wall web between two holes (screw holes included)
M14m	OPEN	3.00	+3.65	+3.05	+2.45	tightest gasket band from a wall hole to the plate edge (screw holes included)
M14n	MET	1.00	+27.69	+26.61	+25.93	nearest bearing face inside to the X 0 rib (if this wall carries it)
M14o	MET	1.00	+11.31	+10.25	+9.19	connector plate top below the outer rim flange
M14p	MET	0.00	+0.90	+0.47	+0.47	screw washers (bonded, 10 across) fully on the plate (edge distance)
M14c	MET	1.00	+8.75	+5.93	+3.22	D: right-angle (down) USB-A plug body, <= 20.0 from the 233-370 rear face, to B16's edge
M14d	MET	1.00	+22.30	+20.06	+17.92	C: shore DC cores (turned down within 12 of the rear) to B16's edge
M14e	MET	1.00	+2.80	+1.73	+1.43	A: patch plug (straight or right-angle, +/-8) under B16 and U51
M14f	MET	1.00	+7.69	+4.87	+2.16	A: patch plug end (<= 40.0 from the rear face) outboard of A22's edge
M14g	MET	1.00	+7.30	+6.17	+5.87	B: USB-C lead (7.0, bent at R 28 into the case) under B16 and U51
M14h	MET	1.00	+19.94	+17.70	+15.56	B: the lead's innermost point outboard of A22's edge

TOLERANCES AND STATUS

Dimensions in mm. Kit parts (this sheet): machined positions and outlines +/-0.10; the 3.0 face plate +/-0.13 and the 6.0 entry plates +/-0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +/-0.10; rebate depth and line +/-0.10; spot-face floors +/-0.10.

Peli: the 1450PF sheet 1453-314-000 rev A states +/-0.76 on one-decimal and +/-0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).

Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.

Each part floats on its fixings (A's and D's flanges 0.24 on M3, C's 0.29, B's body 0.30, E's 0.33, F's 0.30, each M4 0.33); M14j and M14k carry those floats and two machined places per pair.

MeshSat field kit V2, Peli 1450 case set

Connector plate on the back wall (C3): six ruled items, gasket and wall holes

MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
3bdb88df9826 v2/ecad/tools/panel1450.py
214e915b9855 v2/vendor/peli/1450/frame_seat.out
4c6047173e4a v2/cad/connector_plate.py

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 3) from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy.

Sheet 3 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

SECTION through one arrestor on its axis, the wall's Y horizontal: the case INSIDE to the LEFT. Tightened on the bench, cap up, ground lug down; one ground lead per plate to stud F outside (GROUNDING-AND-SHIELDS.md, owed). Not to scale for measurement.

end-wall features Y 0, +-79.0 at Z 92.4 (outside)

inner ribs at Y 0, +-76.2 (inside, Z 26.4..84.6)

axis Z 59.0

5G MAIN 5G DIV 5G ANT3 IRIDIUM LORA

31 pitch

220.2 +-0.10 (case Y +-110.1)

48.9 (Z 34.55..83.45)

M10c OPEN (band)

M10 OPEN (heads below skirt)

cavity floor Z 0

WEST WALL, SEEN FROM OUTSIDE: case +Y (the hinge wall) on your LEFT. 7 arrestors, 16.3 through, spot-face 26.0 x 1.5 on the BACK (dashed); M4 tapped through (drill 3.3) at Y +-45, +-100, Z 40.65 / 77.35; dotted: the 27 mm wall hole

end-wall features Y 0, +-79.0 at Z 92.4 (outside)

inner ribs at Y 0, +-76.2 (inside, Z 26.4..84.6)

axis Z 59.0

WIFI P2P B WIFI P2P A SDR GNSS WIFI 2.4 HF VHF

31 pitch

220.2 +-0.10 (case Y +-110.1)

48.9 (Z 34.55..83.45)

M10c OPEN (band)

M10 OPEN (heads below skirt)

cavity floor Z 0

gasket 2.0 closed-cell, 1.5 compressed

end wall 5.34 (DXF: T5), 27 hole saw

RF entry plate 6.0, hole 16.3

O-ring 0.32 +-0.32 (drawn 0.63, not dimensioned)

.47 in 5/8-24 UNEF thread (11.94, reference only)

arrestor body 23 wide (55 x 23 x 31 max overall)

27

16.3

6.0

5.34

nut + lock washer, class 24.0 x 5.0 (M13b, d)

M13c OPEN (nut radial in the 27)

M13 OPEN (thread takes O-ring, web 4.5, nut 5.0: +1.20 worst)

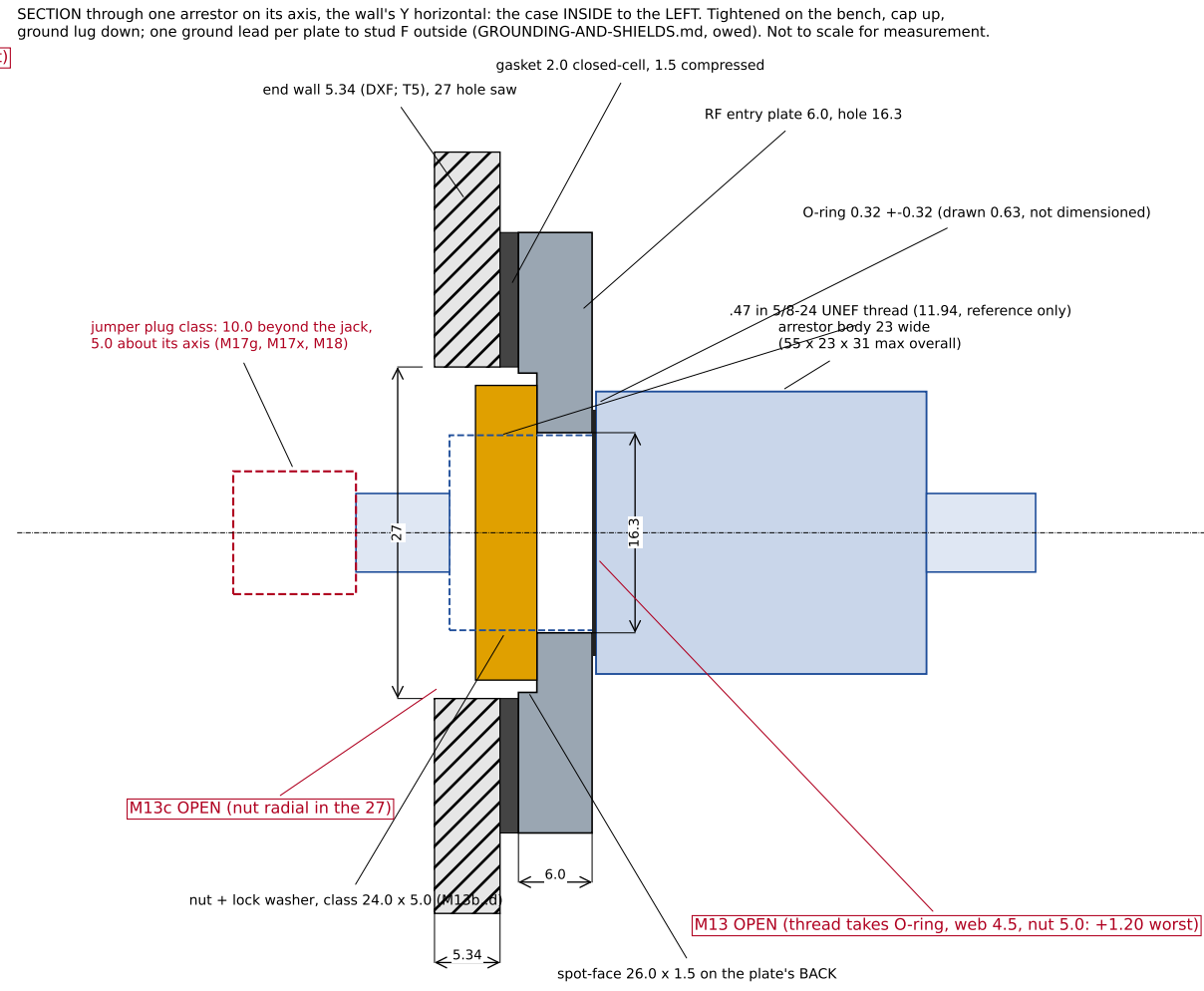
spot-face 26.0 x 1.5 on the plate's BACK

TOLERANCES AND STATUS

Dimensions in mm. Kit parts (this sheet): machined positions and outlines +-0.10; the 3.0 face plate +-0.13 and the 6.0 entry plates +-0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +-0.10; rebate depth and line +-0.10; spot-face floors +-0.10.

Peli: the 1450PF sheet 1453-314-000 rev A states +-0.76 on one-decimal and +-0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED). Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.

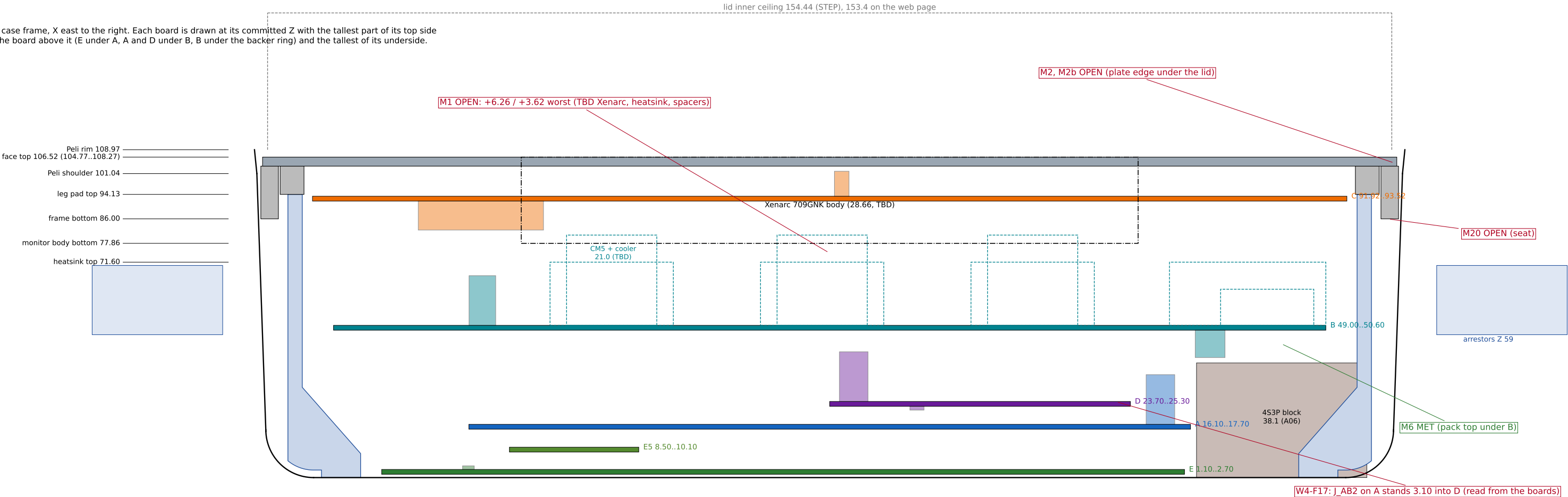
The arrestor's drawing marks every dimension 'for reference only': its .XX +-0.51 is an unstated allowance; its O-ring, nut and washer are drawn, not dimensioned (TBD, PolyPhaser). The jumper plug is a class, not a part: M17g and M17x fail with the class as laid out until the plug is picked (CASE-MARGINS.md 3.4).



TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines +0.10; the 3.0 face plate +0.13 and the 6.0 entry plates +0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +0.10; rebate depth and line +0.10; spot-face floors +0.10.
Pelli: the 1450PF sheet 1453-314-000 rev A states +0.76 on one-decimal and +0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED). Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken into account, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD standard fit, 1:1 fit, or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
The arrestor's drawing marks every dimension 'for reference only': its .XX +0.51 is an unstated allowance; its O-ring, nut and washer are drawn, not dimensioned (TBD, PolyPhaser). The jumper plug is a class, not a part: M17g and M17x fail with the class as laid out until the plug is picked (CASE-MARGINS.md 3.4).

Margin rows this sheet's features enter (frame_seal.out): key, verdict, min, nominal, worst, worst with unstated x2													
M10	OPEN	1.00	+3.65	+1.73	-0.09	highest part inside an end wall (the entry plate's top screw heads, Z 82.35) below the frame skirt	M18	OPEN	1.00	+7.63	+3.36	-0.61	jumper plug's inner end to B16's tall parts in X (tightest: RockBLOCK, which it overlaps in Y and Z)
M11	MET	1.00	+17.89	+17.21	+16.53	inside hardware bearing on an end wall to its ribs at Y 0, +76.2 (the entry plate's screw heads)	M18b	MET	1.00	+8.14	+7.30	+7.30	neighbouring arrestor bodies outside (22.86 across at the 31 pitch)
M11a	MET	2.00	+4.00	+3.40	+2.80	wall web between neighbouring arrestor holes (27 at a 31 pitch)	M18c	MET	1.00	+8.40	+6.80	+6.00	outermost plug (west wall) to the setting legs' inner face in Y (the plug's X range covers the column's)
M11b	MET	2.00	+3.64	+3.04	+2.44	wall web, arrestor hole to entry-plate screw hole (screw -100, 77.35, beside the west wall's -93 site)	M4c	MET	1.00	+5.86	+3.18	+1.80	pack block's east face (X 178.65) to the entry plate's bottom screw heads (inner face X 184.51)
M11c	OPEN	3.00	+3.60	+3.00	+2.40	gasket band from a wall hole to the entry plate's edge (arrestor holes 3.60, 5.0 screw holes 3.60)	M21g	MET	1.00	+4.34	+3.08	+2.20	setting legs' CG (X 180.17) to the entry plate's screw heads at [Y] 100 (X 184.51), in X
M11d	OPEN	0.00	+2.34	+0.40	-0.83	entry-plate screw tips inside the plate's outer face (M4 x 12: nothing stands out under an arrestor)	M17a	OPEN	2.00	+4.44	+2.96	+1.90	east bundle (two R6-316 high, 4.98) between the pack's top (Z 44.58) and the plugs' undersides (Z 54.0)
M11e	OPEN	0.00	+0.51	+0.11	-0.19	each entry-plate M4 (3.98) starts in its 5.0 wall hole: float against hole marking and tapped position	M17b	OPEN	1.00	+2.22	+1.00	+0.20	east cables under the next plug's body: the band's upper layer (the drop itself keeps 8.32)
M11f	MET	0.00	+1.50	+0.92	+0.92	sealing washer's rubber face (8.0 across) covers its 5.0 wall hole, the screw anywhere in its float	M17c	OPEN	0.00	+8.68	+1.56	-2.34	outermost east cable ([Y] 62) and its bundle (R 13.75) inboard of the legs' columns before them, in Y
M11g	OPEN	1.40	+3.66	+1.95	+0.69	entry-plate screws' thread engaged in the plate (at least two pitches, INFERRED)	M17d	OPEN	2.00	+5.42	+3.44	+1.94	east bundle (4.98 wide) in the lane, B16's edge (X 165.0) to the legs' columns (X 175.40): free width
M12	MET	1.00	+4.48	+3.42	+2.36	entry plate's top edge below the end-wall features at Y 0 and +79.0 (Z 87.93)	M17e	OPEN	1.00	+2.85	+1.47	+0.47	east bundle going down beyond the leg: its inner face ([Y] 115.25) to the leg's outer face ([Y] 112.4)
M12b	MET	1.00	+5.05	+4.27	+3.59	entry plate's Y edge on the end wall's flat outer skin (flat to [Y] 115.15 at Z 34.55)	M17f	OPEN	0.00	+2.25	+0.75	-0.75	east front bundle on the floor beside the dock strip's south edge ([Y] 113, gen_pcb.e.py:15)
M13	OPEN	0.00	+2.12	+1.20	+0.38	arrestor nut fully on its thread (.47: 0-ring 0.32, 4.5 web under the spot-face, nut and washer 5.0)	M17g	OPEN, FAILS AS ASSUMED	0.00	-1.21	-2.43	-3.23	east layering under SG MAIN (3 passing): its own cable at its place, clear of the cable below it
M13b	MET	0.00	+3.34	+1.78	+0.52	nut and washer (5.0, from the spot-face floor) inside the wall hole's depth (plate, gasket, wall)	M17x	OPEN, FAILS AS ASSUMED	1.00	+3.89	-0.38	-4.35	east bundle's inboard column under the plugs to B16's edge (X 165.0), axis at the plug's inner end
M13c	OPEN	0.00	+1.50	+0.48	-0.12	nut and washer (24.0) inside the 27 wall hole, radial	M17w	OPEN	1.00	+6.38	+2.11	-1.86	west cables (leaving downward, axis within 4.0 of the plug's end) outboard of B16's edge
M13d	MET	0.00	+1.00	+0.58	+0.58	nut and washer (24.0) inside the 26.0 spot-face, radial							

SECTION at Y 0, case frame, X east to the right. Each board is drawn at its committed Z with the tallest part of its top side that lies under the board above it (E under A, A and D under B, B under the backer ring) and the tallest of its underside.



THE STACK FROM THE FLOOR (panel1450.STACK)
VHB 5952 pads under the dock strip 0.00 .. 1.10
board E dock strip 1.10 .. 2.70
blind-mate gap spacer 2.70 .. 16.10
board A 16.10 .. 17.70
A-to-B bay spacer 17.70 .. 49.00
board B 49.00 .. 50.60
B top copper (B_TOP_Z) 50.60
D on 6.0 standoffs 23.70 .. 25.30
E5 dock block (face 7.4 over E) 8.50 .. 10.10
C backer ring (10 under the plate) 91.92 .. 93.52
face plate on the frame ring 103.52 .. 106.52
P: Z not set until the pack hold-down (S-27)

BAYS READ FROM THE BOARDS (gap - lower part - upper part where they overlap in plan)
A to B outside D (bay): gap 31.30
pair J_AB1 9.10 under J_AB1 9.10: +13.10
tallest single J_54V 16.50 (lower top): leaves +14.80
A to D (mezzanine standoffs): gap 6.00
pair J_AB2 9.10 under C18 0.80: -3.90
tallest single J_AB2 9.10 (lower top): leaves -3.10
B to C backer ring: gap 41.32
pair J_PANEL 9.10 under J_PANEL 9.64: +22.58

tallest single J_54V 16.50 (lower top): leaves +24.82
D to B: gap 23.70
pair J_ANT 9.55 under C286 0.80: +13.35
tallest single J_PWR1 16.50 (lower top): leaves +7.20
E to A (blind-mate gap): gap 13.40
tallest single Q10 1.20 (lower top): leaves +12.20
E5 to A (dock contacts): gap 6.00
tallest single H1 0.00 (lower top): leaves +6.00
floor under E (E's underside parts on the VHB pads): gap 1.10

B's parts under the monitor (body bottom 77.86):
J_5V_S1 h 16.50 leaves +10.76
J_5V_S2 h 16.50 leaves +10.76
J_5V_S3 h 16.50 leaves +10.76
J_RB9704 h 9.10 leaves +18.16
B's underside over the pack pocket: J_AB2 9.10,
lowest Z 39.90

Heights: each footprint's own KiCad 9 library model (kicad-packages3D 9.0.9); SUBSTITUTE where the named model file is missing; CLASS or MAKER where the footprint names none (zstack.py HEIGHT_CLASSES).

Margin rows this sheet's features enter (frame_seat.out): key, verdict, min, nominal, worst, worst with unstated x2

M1	OPEN	2.00	+6.26	+3.62	+1.80	monitor body over the CM5 heatsinks
M2	OPEN	1.00	+4.45	+1.84	-0.44	rebated plate band below the rim, under the lid's wall
M2b	OPEN	1.00	+2.94	+1.25	+0.29	full-thickness face (368.0 x 253.0) inside the lid's opening, in X (Y: 3.29 nominal)
M3	OPEN	1.00	+19.92	+16.39	+14.11	space under the QMX tray for the face parts (lid STEP 45.47; web 44.45 in the worst)
M6	MET	1.00	+4.42	+3.66	+2.90	pack top under B16's underside (A06 3.32 at its pessimistic base, plus the 1.1 VHB lift)
M8z	OPEN	0.00	+2.48	+0.10	-2.18	plate underside above the shoulder, so the plate edge faces the rim zone
M20	OPEN					the frame's seat: bottom Z 86.00 nominal, 84.38..87.62 on the legs (T2)
M4a	OPEN	1.00	+3.71	+1.85	+0.85	pack block's east corner to Peli's R 15.88 floor fillet (west face X 122, A06)
M5	OPEN	1.00	+3.65	+1.77	+0.27	pack group (205.5 long, A06) to the east legs' inner faces at Y 106.4, per side
M15a	MET	1.00	+7.30	+5.92	+4.54	dock strip's underside (on 1.1 pads) to the front floor fillet (edge Y -113, gen_pcb_e.py:15)
M15b	OPEN	0.00	+1.49	+0.11	-1.27	dock strip's corner pads on the flat floor (edge to the tangent)

TOLERANCES AND STATUS

Dimensions in mm. Kit parts (this sheet): machined positions and outlines ± 0.10 ; the 3.0 face plate ± 0.13 and the 6.0 entry plates ± 0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height ± 0.10 ; rebate depth and line ± 0.10 ; spot-face floors ± 0.10 .
Peli: the 1450PF sheet 1453-314-000 rev A states ± 0.76 on one-decimal and ± 0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
Stack: JLC 1.6 ± 0.16 per board and 3M VHB 5952 1.1 $\pm 10\%$ (VERIFIED); the gap and bay spacers are unnamed (TBD); B bow 0.3 (INFERRED). The section is at Y 0 with each board's tallest part drawn at its own X.

MeshSat field kit V2, Peli 1450 case set

The Z stack, floor to lid: boards at their committed Z, the face on the legs, the bays read from the boards

MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

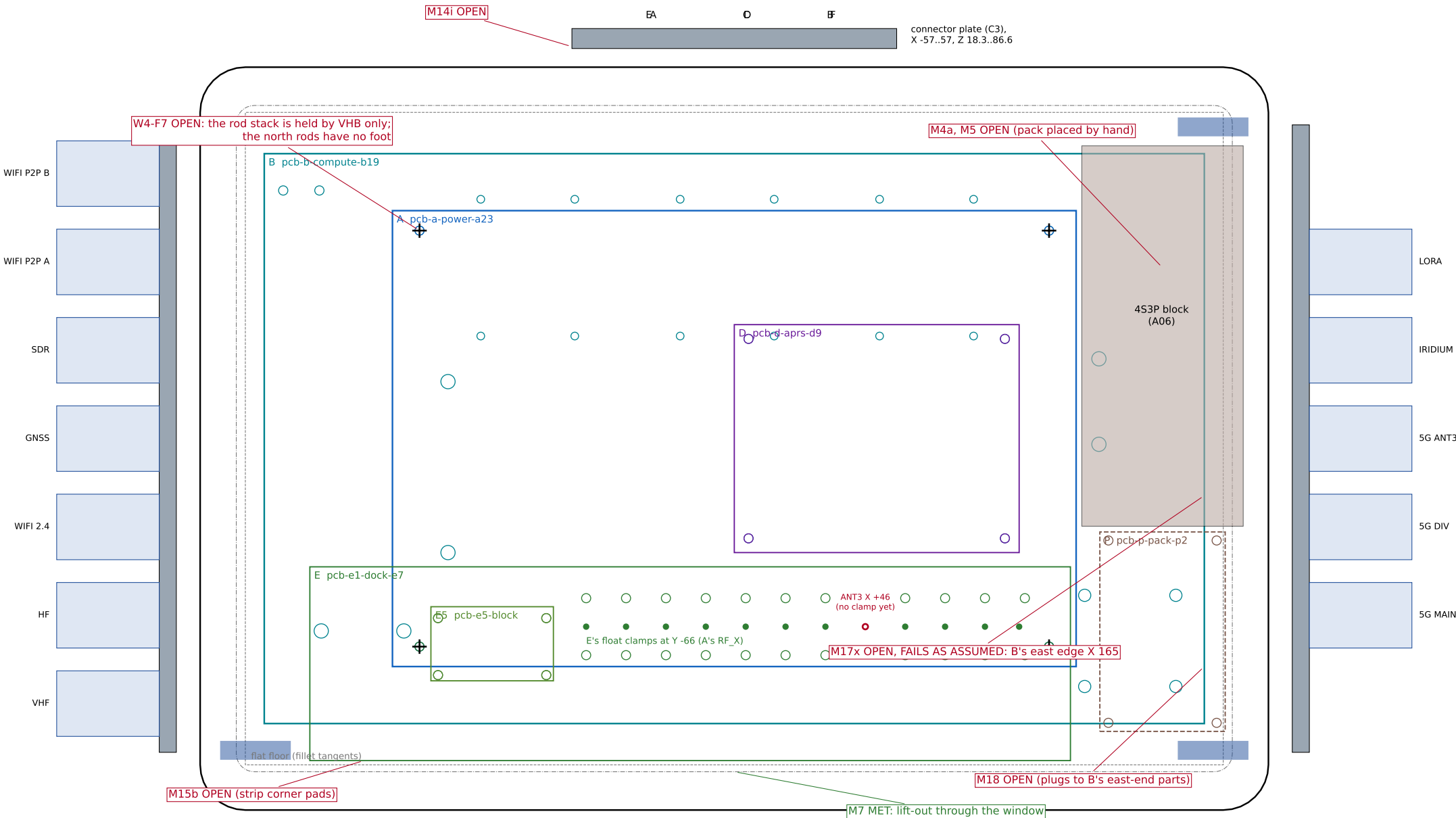
Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
3bdbc88df9826 v2/ecad/tools/panel1450.py
214e915b9855 v2/vendor/peli/1450/frame_seat.out
b4fb79d98d2f v2/cad/zstack.json
1ba75de7ceef v2/cad/zstack.py
58e26c67987b v2/ecad/pcb-a-power-a23/pcb-a-power.kicad_pcb
2e64b5bf2d9c v2/ecad/pcb-b-compute-b19/pcb-b-compute.kicad_pcb

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 5) from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy.

Sheet 5 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

PLAN, case frame, seen from above: inner wall at the fillet tangent Z 15.32 (375.01 x 260.71), flat floor dashed, frame window dash-dot; boards at their committed outlines with their mounting holes; legs blue; rods +; arrestors outside the end walls; the back wall at the top.



MOUNTING (read from the committed boards)
E: 28 holes >= 2.0: 3.2
E5: 4 holes >= 2.0: 3.2
A: 8 holes >= 2.0: 3.2
D: 4 holes >= 2.0: 3.2
B: 30 holes >= 2.0: 2.0, 2.7, 3.2, 4.3, 5.0
C: 12 holes >= 2.0: 15.0, 16.2, 17.0, 19.2, 2.2, 28.6
P: 4 holes >= 2.0: 3.2

RETENTION (what holds what)
- face plate: ten 6-32 x 1/2 in into Peli's inserts (C1)
- frame: Peli's four self-tapping screws; the legs in compression
- rod stack (E, A, B on four M3 rods at +-110.5, +-73): the dock strip's VHB 5952 pads only; the north rods have no foot (W4-F7, OPEN; proposal RF-1 not ruled; T8 bond test)
- D on A: four M3 x 6 standoffs; C under the plate: eight PEM S0-M3-10 standoffs; E5 on E: M3 standoffs
- pack group: NOT DESIGNED (S-27, CON-006); placed by hand in this drawing (M4a, M5 OPEN)
- RockBLOCK 9704: four M4 on its bracket (H17..H20); fastener type and drop owed (ASSEMBLY.md)

ACCESS
- lift-out: plate off, frame and legs stay; the stack clears the window by 8.25 per side at the worst (M7 MET)
- back wall: six items on the connector plate (C3); four picks OPEN
- end walls: twelve arrestors on two entry plates (C2, C4), jumpers to E's clamps on planned routes (every jumper row OPEN)
- service ports on B (J FLASH1..3, USB-C) and headers reachable only with the plate off

Margin rows this sheet's features enter (frame_seat.out): key, verdict, min, nominal, worst, worst with unstated x2

M4a	OPEN	1.00	+3.71	+1.85	+0.85	pack block's east corner to Peli's R 15.88 floor fillet (west face X 122, A06)
M4b	MET	1.00	+9.68	+7.38	+6.38	pack block's east face to the east wall
M5	OPEN	1.00	+3.65	+1.77	+0.27	pack group (205.5 long, A06) to the east legs' inner faces at Y 106.4, per side
M7	MET	1.00	+9.83	+8.25	+7.05	stack lift-out through the frame window, per side in X (Y has 16.91)
M15a	MET	1.00	+7.30	+5.92	+4.54	dock strip's underside (on 1.1 pads) to the front floor fillet (edge Y -113, gen_pcb_e.py:15)
M15b	OPEN	0.00	+1.49	+0.11	-1.27	dock strip's corner pads on the flat floor (edge to the tangent)
M16	MET	1.00	+41.49	+40.11	+38.73	rods (+-110.5, +-73) to the long-wall fillet tangents (61.14 to the end walls)
M17x	OPEN, FAILS AS ASSUMED	1.00	+3.89	-0.38	-4.35	east bundle's inboard column under the plugs to B16's edge (X 165.0), axis at the plug's inner end
M18	OPEN	1.00	+7.63	+3.36	-0.61	jumper plug's inner end to B16's tall parts in X (tightest: RockBLOCK, which it overlaps in Y and Z)
M14i	OPEN	1.00	+1.93	+1.15	+0.47	connector plate edge to the hinge fairings' bases in X
M21f	MET	1.00	+6.40	+4.52	+3.02	leg to B16's corner (165, 100) in Y

TOLERANCES AND STATUS

Dimensions in mm. Kit parts (this sheet): machined positions and outlines +-0.10; the 3.0 face plate +-0.13 and the 6.0 entry plates +-0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +-0.10; rebate depth and line +-0.10; spot-face floors +-0.10.
Peli: the 1450PF sheet 1453-314-000 rev A states +-0.76 on one-decimal and +-0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.

MeshSat field kit V2, Peli 1450 case set

The case in plan: boards, rods, legs, pack group, wall plates, jacks and dock clamps

MESH-SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

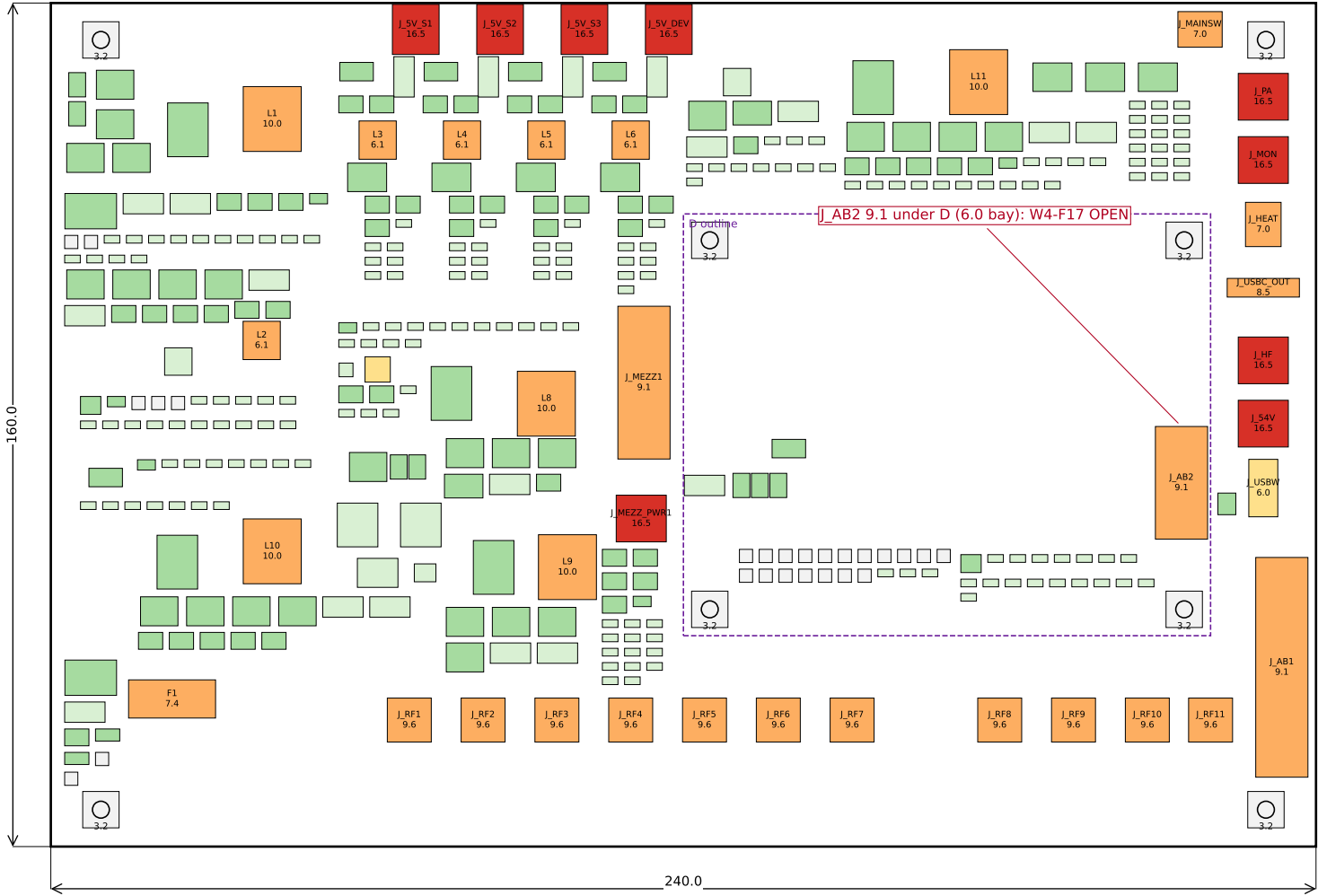
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214e915b9855 v2/vendor/peli/1450/frame_seat.out
b4fb79d90d2f v2/cad/zstack.json

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 6) from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy.

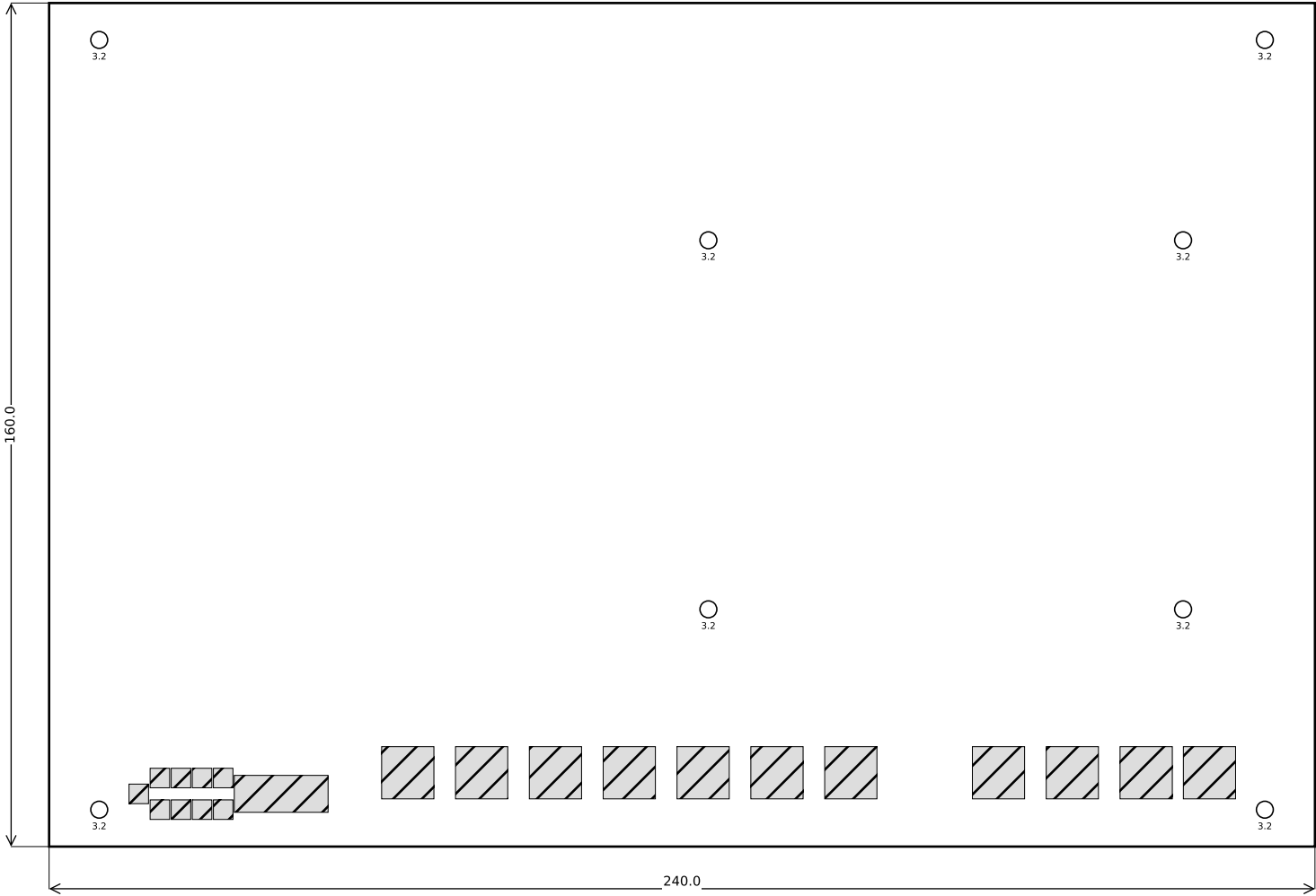
Sheet 6 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

TOP SIDE (seen from above)



BOTTOM SIDE (seen from above, through the board: not mirrored)



v2/ecad/pcb-a-power-a23/pcb-a-power.kicad_pcb sha256 58e26c67987b1daa (gen_pcb_a.py 0X, 0Y; on the blind-mate gap spacer)
outline X -120.00..120.00, Y -80.00..80.00 (case frame), thickness 1.60 (the file's), Z 16.10..17.70
tallest top: J_HF 16.50 (JST_VH_B2P-VH_1x02_P3.96mm Vertical); tallest bottom: none
parts 400 by height source: BODYLESS 33, CLASS 31, MAKER 9, MATES 21, MODEL 306
holes: H1 3.2, H3 3.2, H5 3.2, H7 3.2, H6 3.2, H8 3.2, H2 3.2, H4 3.2
colour by height above the board surface: no body, <=1, <=3, <=6, <=10, >10 mm (light grey, greens, yellow, orange, red); grey hatched: a height
another record owns (COVERED by a module envelope, a PANEL part hanging from the plate, a contact that MATES across a bay); magenta cross-hatched:
TBD; B's module envelopes dashed.

HOLES 8 of 8 (drill 2.0 or more), centre in the case frame, mm				
ref	X	Y	drill	what
H1	-110.50	-73.00	3.20	M3 rod R1
H3	-110.50	73.00	3.20	M3 rod R3
H5	5.00	-35.00	3.20	M3 standoff, mezzanine
H7	5.00	35.00	3.20	M3 standoff, mezzanine
H6	95.00	-35.00	3.20	M3 standoff, mezzanine
H8	95.00	35.00	3.20	M3 standoff, mezzanine
H2	110.50	-73.00	3.20	M3 rod R2
H4	110.50	73.00	3.20	M3 rod R4

TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines ± 0.10 ; the 3.0 face plate ± 0.13 and the 6.0 entry plates ± 0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height ± 0.10 ; rebate depth and line ± 0.10 ; spot-face floors ± 0.10 .
Peli: the 1450PF sheet 1453-314-000 rev A states ± 0.76 on one-decimal and ± 0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
Heights are nominal library bodies at the committed placements; a board's plan tolerance is JLC's outline class (± 0.2 , not held) and the stack's placement ± 1.0 by hand (INFERRED).

MeshSat field kit V2, Peli 1450 case set

Board envelope A power and I/O: outline, holes and parts by height on both sides, read from the committed board

MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

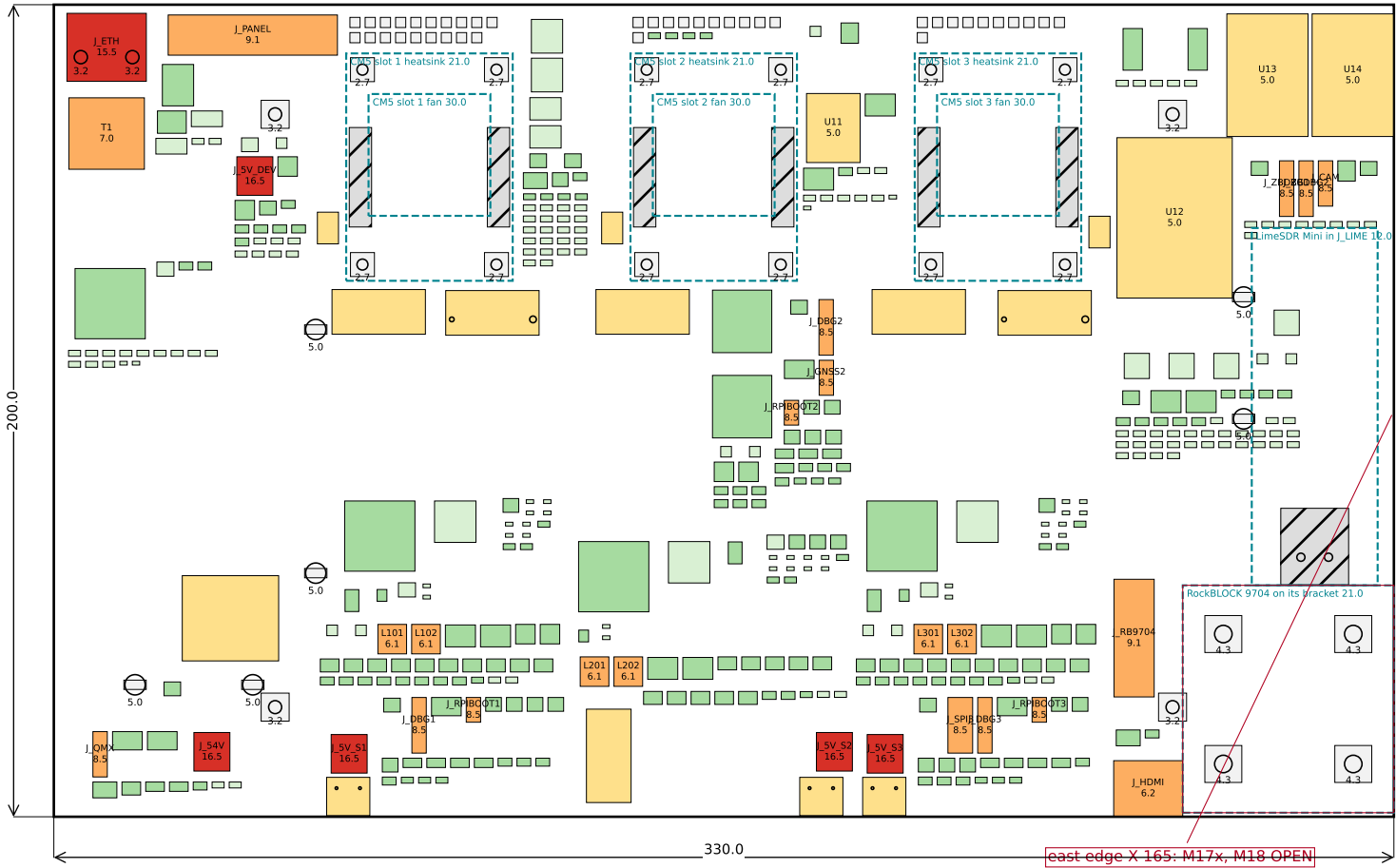
Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
b4fb79d98d2f v2/cad/zstack.json
1ba75de7ceef v2/cad/zstack.py
58e26c67987b v2/ecad/pcb-a-power-a23/pcb-a-power.kicad_pcb
3fad2fb95f58 v2/cad/zstack-models.json

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 7) from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy.

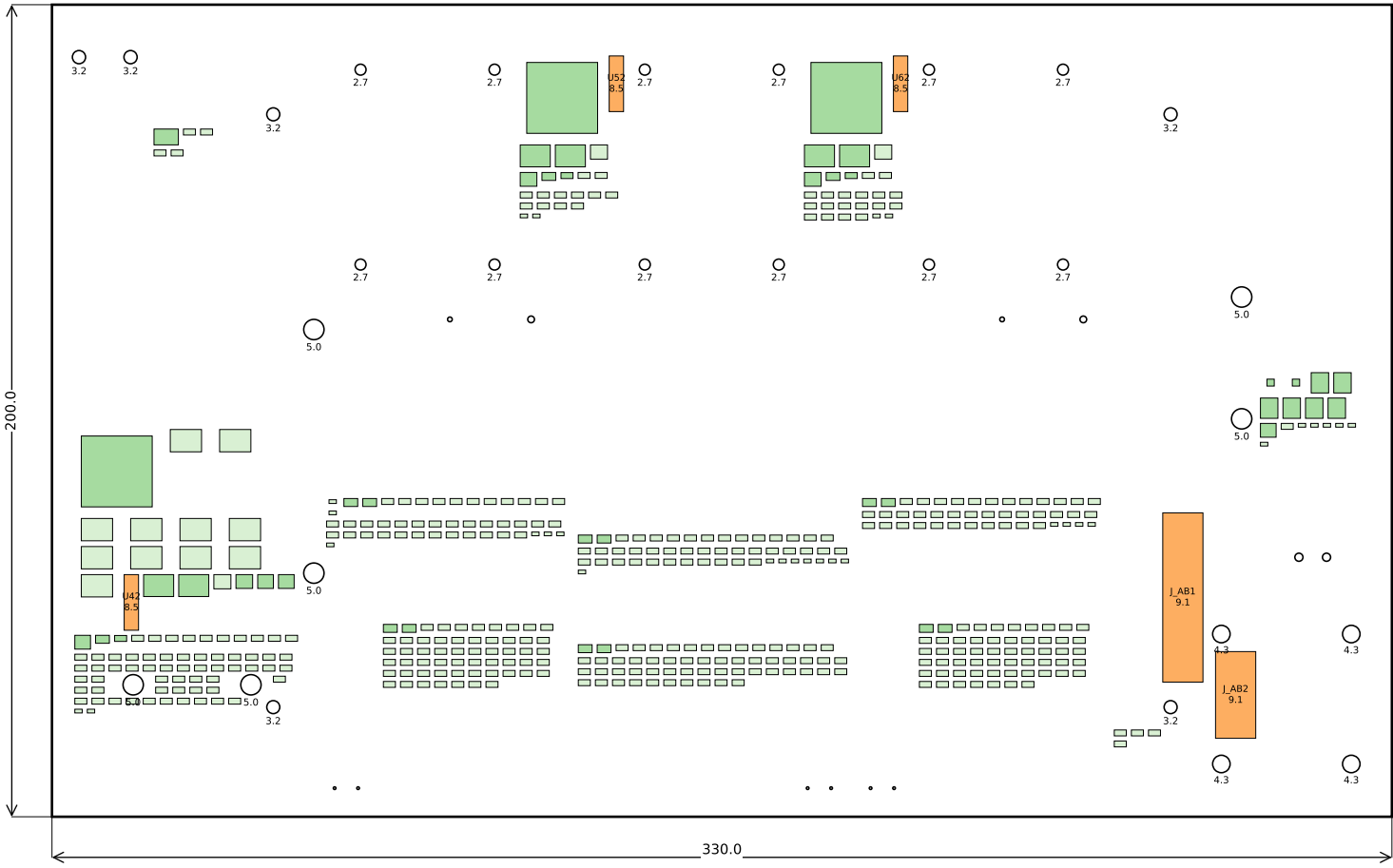
Sheet 7 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

TOP SIDE (seen from above)



BOTTOM SIDE (seen from above, through the board: not mirrored)



v2/ecad/pcb-b-compute-b19/pcb-b-compute.kicad_pcb sha256 2e64b5bf2d9cd3bc (gen_pcb_b.py OX, OY; on the A-to-B bay spacer)
outline X -165.00..165.00, Y -100.00..100.00 (case frame), thickness 1.60 (the file's), Z 49.00..50.60
tallest top: J_5V_DEV 16.50 (JST_VH_B2P-VH_1x02_P3.96mm_Vertical); tallest bottom: J_AB2 9.10 (IDC-Header_2x05_P2.54mm_Vertical_NarrowPad)
parts 957 by height source: BODYLESS 67, CLASS 27, COVERED 7, MAKER 8, MODEL 848
holes: J_ETH 3.2, J_ETH 3.2, S_QMX3 5.0, S_QMX4 5.0, H1 3.2, H3 3.2, S_QMX2 5.0, S_QMX1 5.0, H5 2.7, H6 2.7, H7 2.7, H8 2.7, H9 2.7, H10 2.7, H11 2.7, H12 2.7, H13 2.7, H14 2.7, H15 2.7, H16 2.7, H2 3.2, H4 3.2, H17 4.3, H18 4.3, S_LIME1 5.0, S_LIME2 5.0, J_LIME 2.0, J_LIME 2.0, H19 4.3, H20 4.3
colour by height above the board surface: no body, <=1, <=3, <=6, <=10, >10 mm (light grey, greens, yellow, orange, red); grey hatched: a height another record owns (COVERED by a module envelope, a PANEL part hanging from the plate, a contact that MATES across a bay); magenta cross-hatched: TBD; B's module envelopes dashed.

HOLES 30 of 40 (drill 2.0 or more), centre in the case frame, mm									
ref	X	Y	drill	what	ref	X	Y	drill	what
J_ETH	-158.35	87.00	3.25	RJ45 jack (Amphenol RJHSE5380): patch lead to the seale	H15	84.00	36.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S3
J_ETH	-145.65	87.00	3.25	RJ45 jack (Amphenol RJHSE5380): patch lead to the seale	H16	84.00	84.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S3
S_QMX3	-145.00	-67.50	5.00	tie slot	H2	110.50	-73.00	3.20	M3 rod R2
S_QMX4	-116.00	-67.50	5.00	tie slot	H4	110.50	73.00	3.20	M3 rod R4
H1	-110.50	-73.00	3.20	M3 rod R1	H17	123.00	-87.00	4.30	M4, GC 9704 bracket
H3	-110.50	73.00	3.20	M3 rod R3	H18	123.00	-55.00	4.30	M4, GC 9704 bracket
S_QMX2	-100.50	-40.00	5.00	tie slot	S_LIME1	128.00	-2.00	5.00	tie slot
S_QMX1	-100.50	20.00	5.00	tie slot	S_LIME2	128.00	28.00	5.00	tie slot
H5	-89.00	36.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S1	J_LIME	142.10	-36.08	2.00	USB 3.0 type A receptacle (Wuerth 692122030100 land): t
H6	-89.00	84.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S1	J_LIME	148.90	-36.08	2.00	USB 3.0 type A receptacle (Wuerth 692122030100 land): t
H7	-56.00	36.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S1	H19	155.00	-87.00	4.30	M4, GC 9704 bracket
H8	-56.00	84.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S1	H20	155.00	-55.00	4.30	M4, GC 9704 bracket
H9	-19.00	36.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S2					
H10	-19.00	84.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S2					
H11	14.00	36.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S2					
H12	14.00	84.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S2					
H13	51.00	36.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S3					
H14	51.00	84.00	2.70	M2.5 standoff 4.0 mm, CM5 slot S3					

TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines ± 0.10 ; the 3.0 face plate ± 0.13 and the 6.0 entry plates ± 0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height ± 0.10 ; rebate depth and line ± 0.10 ; spot-face floors ± 0.10 .
Peli: the 1450PF sheet 1453-314-000 rev A states ± 0.76 on one-decimal and ± 0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
Heights are nominal library bodies at the committed placements; a board's plan tolerance is JLC's outline class (± 0.2 , not held) and the stack's placement ± 1.0 by hand (INFERRED).

MeshSat field kit V2, Peli 1450 case set

Board envelope B compute: outline, holes and parts by height on both sides, read from the committed board

MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

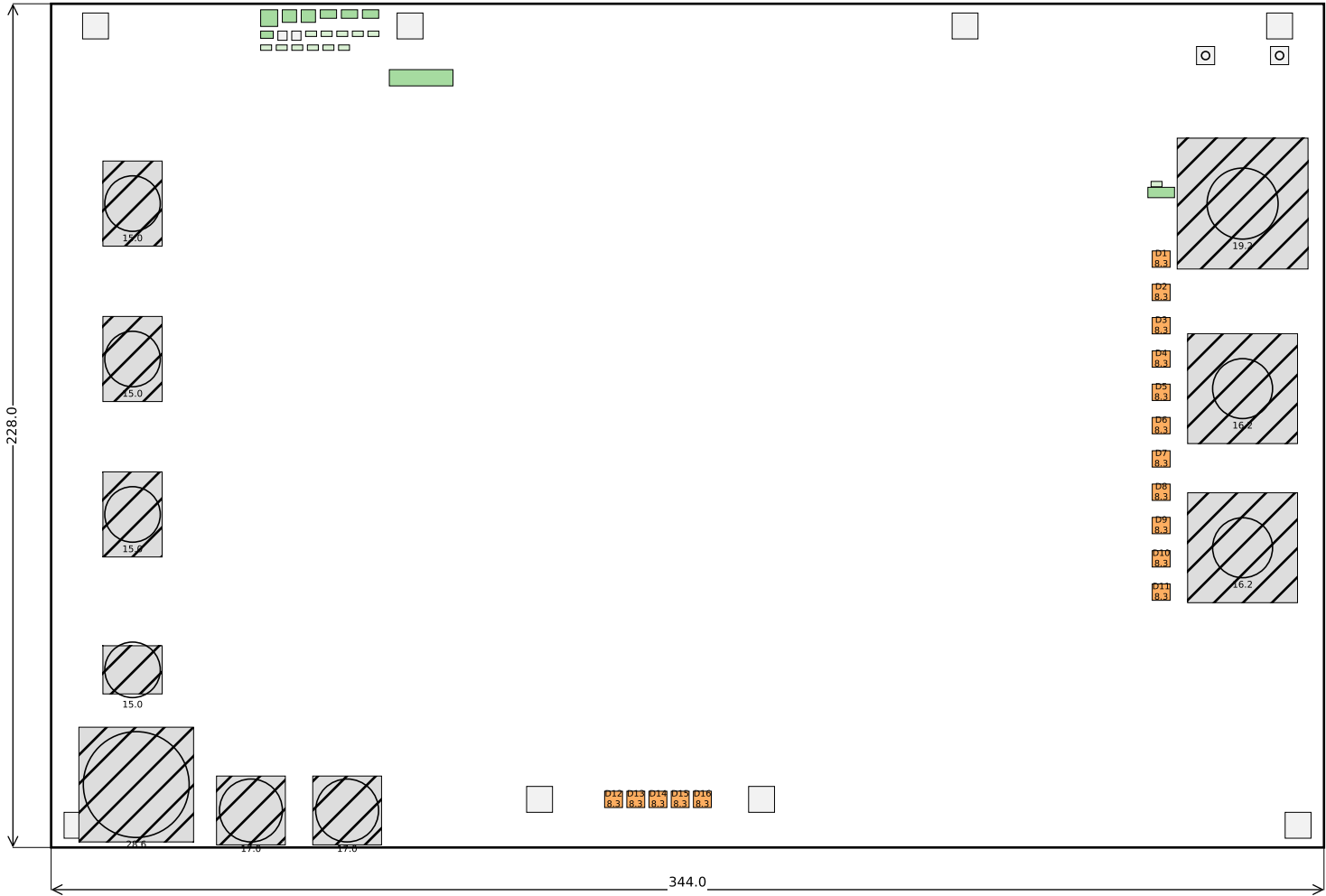
Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
b4fb79d90d2f v2/cad/zstack.json
1ba75de7ceef v2/cad/zstack.py
2e64b5bf2d9c v2/ecad/pcb-b-compute-b19/pcb-b-compute.kicad_pcb
3fad2fb95f58 v2/cad/zstack-models.json

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 8) from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy.

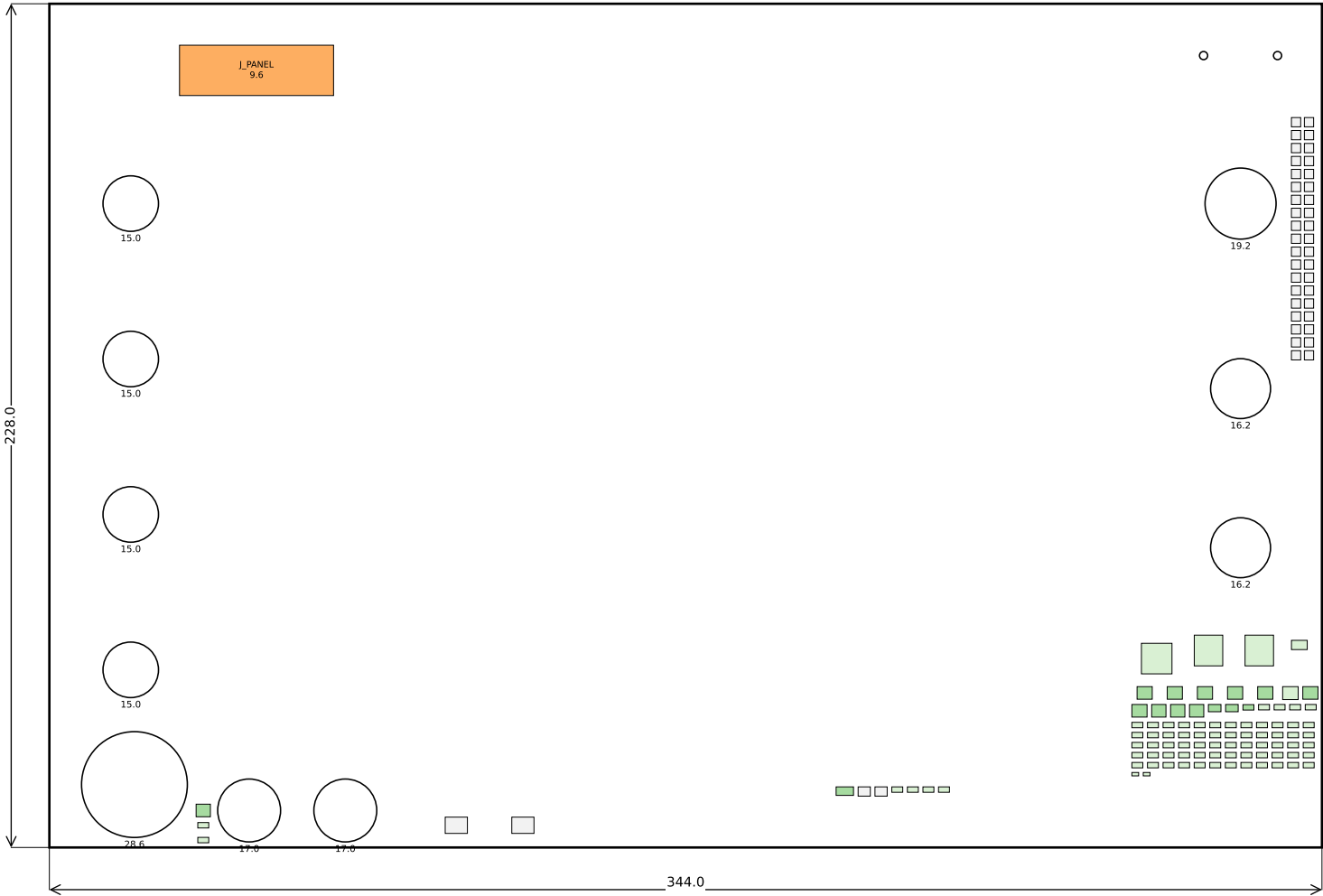
Sheet 8 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

TOP SIDE (seen from above)



BOTTOM SIDE (seen from above, through the board: not mirrored)



v2/ecad/pcb-c-display-c8/pcb-c-display.kicad_pcb sha256 2a273803757c68fb (gen_pcb_c.py 0X, 0Y; 10 mm standoffs under the face plate)
outline X -172.00..172.00, Y -114.00..114.00 (case frame), thickness 1.60 (the file's), Z 91.92..93.52
tallest top: D16 8.30 (LED_D3.0mm); tallest bottom: J_PANEL 9.64 (IDC-Header_2x13_P2.54mm_Vertical_SMD)
parts 194 by height source: BODYLESS 54, CLASS 2, MODEL 128, PANEL 10
holes: SW_LIGHT 15.0, SW_ZERO 15.0, SW_EMCON 15.0, SW_SOS 15.0, BZ1 28.6, J_HSJ1 17.0, J_HSJ2 17.0, CAM_H1 2.2, SW_TEST 16.2, SW_PI 16.2, SW_MAIN 19.2, CAM_H2 2.2
colour by height above the board surface: no body, <=1, <=3, <=6, <=10, >10 mm (light grey, greens, yellow, orange, red); grey hatched: a height another record owns (COVERED by a module envelope, a PANEL part hanging from the plate, a contact that MATES across a bay); magenta cross-hatched: TBD; B's module envelopes dashed.

HOLES 12 of 12 (drill 2.0 or more), centre in the case frame, mm				
ref	X	Y	drill	what
SW_LIGHT	-150.00	-66.00	15.00	LIGHTING DAY/NIGHT/BLACKOUT toggle DPDT ON-ON-ON (pole 1: rail, pole 2: sense); NKK M2044SD3A01 on the D3 splashproof bushing, its 0-ring (spare AT516
SW_ZERO	-150.00	-24.00	15.00	ZEROIZE locking toggle, maintained (APEM 5636ADKB-2V, hinged safety cover; ruling 32.13); K front seal in the keyed 6.5 hole; ZEROIZE HW low = wipe th
SW_EMCON	-150.00	18.00	15.00	EMCON locking toggle (closed = TX inhibit, a hardware line: TX INHIBIT n low and EMCON HW low, the same sense; APEM 5636ADKB-2V, hinged safety cover);
SW_SOS	-150.00	60.00	15.00	SOS locking toggle, maintained (APEM 5636ADKB-2V, both positions locked, red boot, hinged safety cover per 32.50; ruling 32.13); K front seal in the k
BZ1	-149.00	-97.00	28.60	IP67 panel-mount sounder, 5 V DC continuous, bezel gasket on the face, two flying leads (Floyd Bell MC-09-530-0 class)
J_HSJ1	-118.00	-104.00	17.00	U-174/U headset jack 1 (Amphenol Nexus class, drawing owed): 17 mm hole through the backer, bushing nut below; its five leads run to D8 J_HS1
J_HSJ2	-92.00	-104.00	17.00	U-174/U headset jack 2 (Amphenol Nexus class, drawing owed): 17 mm hole through the backer, bushing nut below; its five leads run to D8 J_HS2
CAM_H1	140.00	100.00	2.20	M2 screw of the USB camera module (25 x 25 board, sheet owed) behind the plate's sealed window
SW_TEST	-33.00	-33.00	16.20	TEST/ACK 16 mm momentary, white ring; C&K ATP16-SL1-203-M05A-04G; silicone gasket washer under the bezel
SW_PI	150.00	10.00	16.20	PI 16 mm recessed momentary, amber ring (PI SHDN REQ to the modules through the controller); C&K ATP16-SL1-403-M05A-04G; silicone gasket washer under
SW_MAIN	150.00	60.00	19.20	MAIN PWR 19 mm momentary, green ring (to A22 J_MAINSW); C&K ATP19-SL1-603-B05A-03G; silicone gasket washer under the bezel
CAM_H2	160.00	100.00	2.20	M2 screw of the USB camera module (25 x 25 board, sheet owed) behind the plate's sealed window

TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines +/-0.10; the 3.0 face plate +/-0.13 and the 6.0 entry plates +/-0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +/-0.10; rebate depth and line +/-0.10; spot-face floors +/-0.10.
Peli: the 1450PF sheet 1453-314-000 rev A states +/-0.76 on one-decimal and +/-0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
Heights are nominal library bodies at the committed placements; a board's plan tolerance is JLC's outline class (+/-0.2, not held) and the stack's placement +/-1.0 by hand (INFERRED).

MeshSat field kit V2, Peli 1450 case set

Board envelope C panel backer ring: outline, holes and parts by height on both sides, read from the committed board

MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

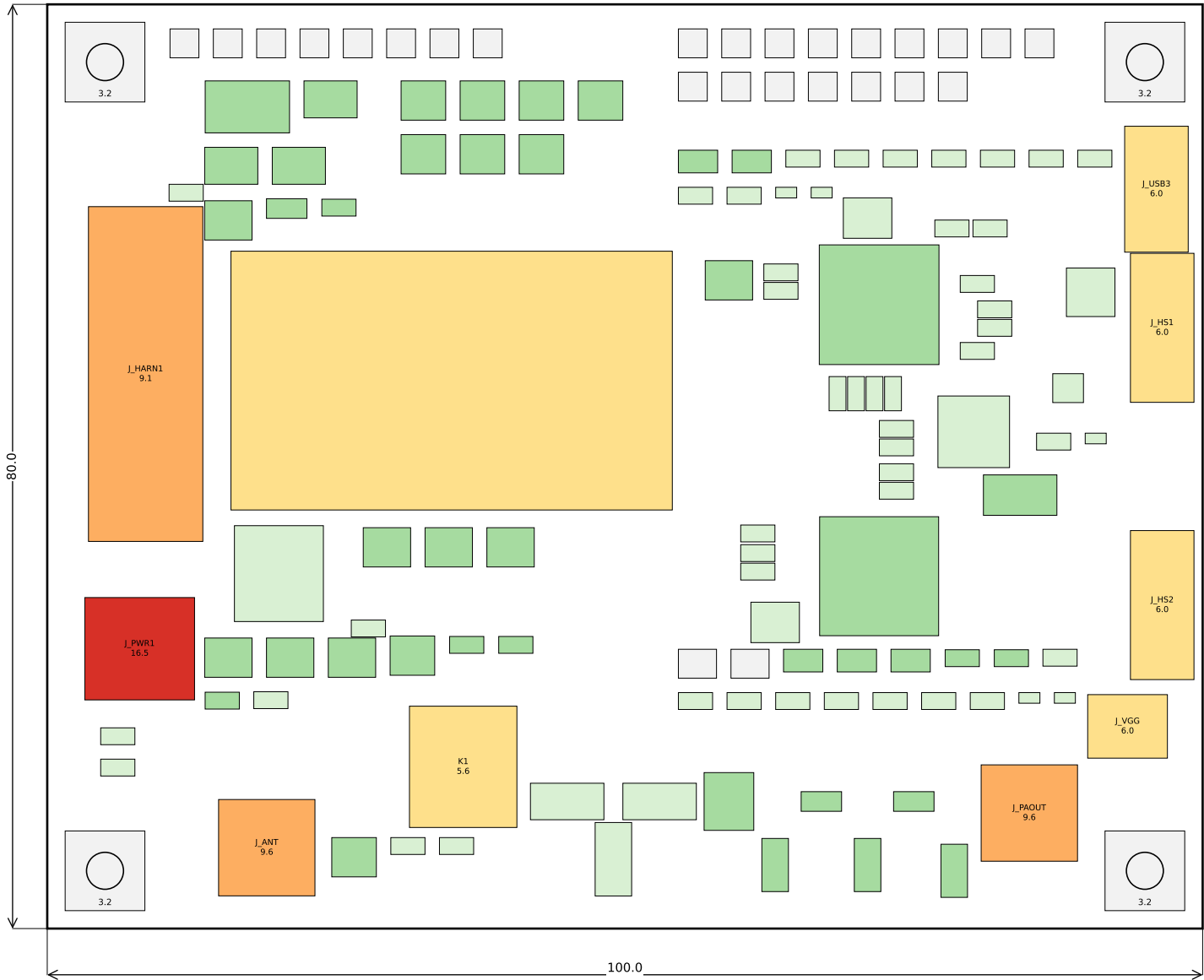
Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
b4fb79d90d2f v2/cad/zstack.json
1ba75de7ceef v2/cad/zstack.py
2a273803757c v2/ecad/pcb-c-display-c8/pcb-c-display.kicad_pcb
3fad2fb95f58 v2/cad/zstack-models.json

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 9) from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy.

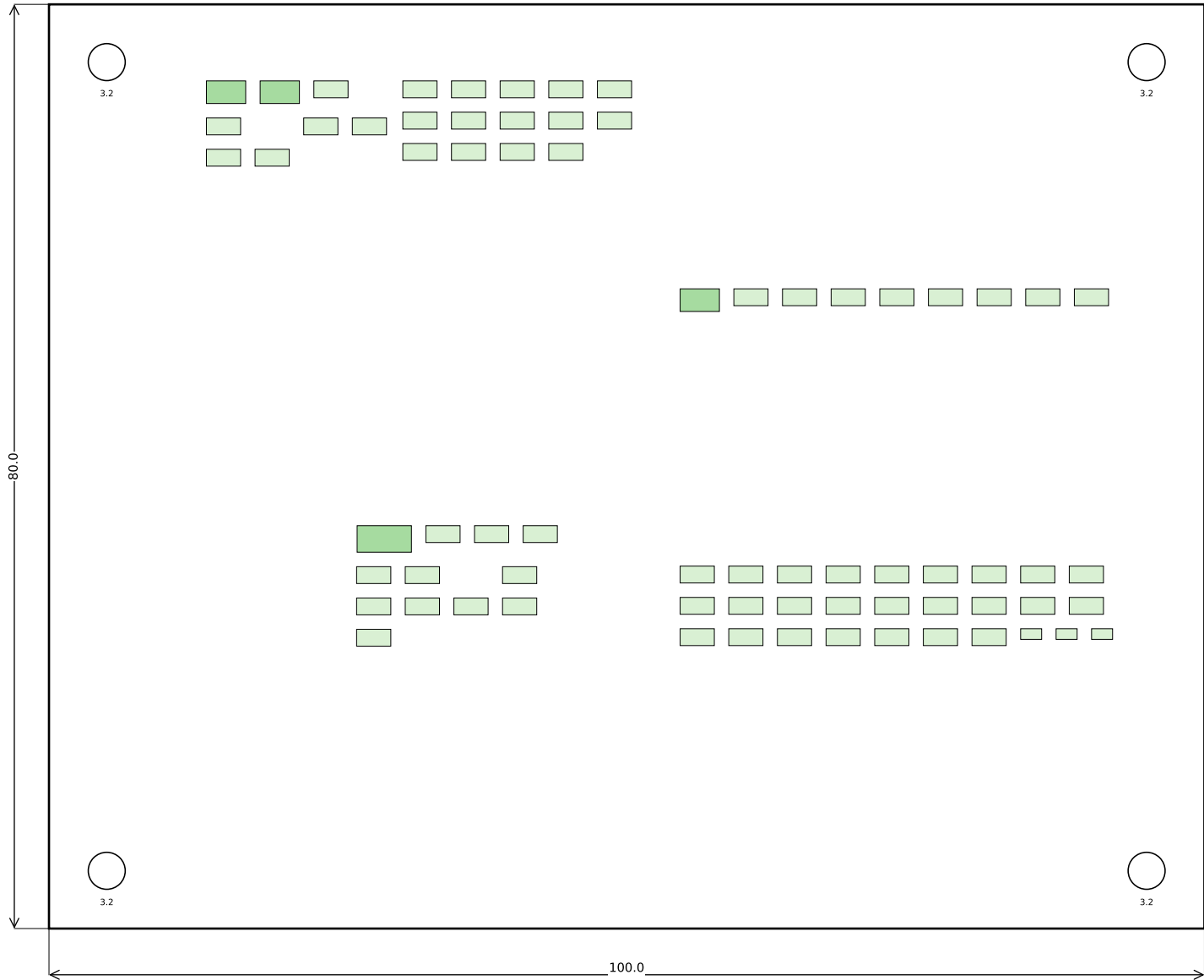
Sheet 9 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

TOP SIDE (seen from above)



BOTTOM SIDE (seen from above, through the board: not mirrored)



v2/ecad/pcb-d-aprs-d9/pcb-d-aprs.kicad_pcb sha256 929bf82d2bf6eed4 (gen_pcb_d.py OX, OY; its centre at board A's MEZZ_RECT centre (gen_pcb_a.py:51), on 6 mm standoffs)
outline X 0.00..100.00, Y -40.00..40.00 (case frame), thickness 1.60 (the file's), Z 23.70..25.30
tallest top: J_PWR1 16.50 (JST_VH_B2P-VH_1x02_P3.96mm_Verical); tallest bottom: D2 1.26 (D_SOD-123)
parts 211 by height source: BODYLESS 30, CLASS 1, MAKER 1, MODEL 179
holes: H1 3.2, H3 3.2, H2 3.2, H4 3.2
colour by height above the board surface: no body, <=1, <=3, <=6, <=10, >10 mm (light grey, greens, yellow, orange, red); grey hatched: a height another record owns (COVERED by a module envelope, a PANEL part hanging from the plate, a contact that MATES across a bay); magenta cross-hatched: TBD; B's module envelopes dashed.

HOLES 4 of 4 (drill 2.0 or more), centre in the case frame, mm				
ref	X	Y	drill	what
H1	5.00	-35.00	3.20	M3 standoff on A22 (5/95, +-35)
H3	5.00	35.00	3.20	M3 standoff on A22 (5/95, +-35)
H2	95.00	-35.00	3.20	M3 standoff on A22 (5/95, +-35)
H4	95.00	35.00	3.20	M3 standoff on A22 (5/95, +-35)

TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines +-0.10; the 3.0 face plate +-0.13 and the 6.0 entry plates +-0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +-0.10; rebate depth and line +-0.10; spot-face floors +-0.10.
Peli: the 1450PF sheet 1453-314-000 rev A states +-0.76 on one-decimal and +-0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
Heights are nominal library bodies at the committed placements; a board's plan tolerance is JLC's outline class (+-0.2, not held) and the stack's placement +-1.0 by hand (INFERRED).

MeshSat field kit V2, Peli 1450 case set

Board envelope D APRS mezzanine: outline, holes and parts by height on both sides, read from the committed board

MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

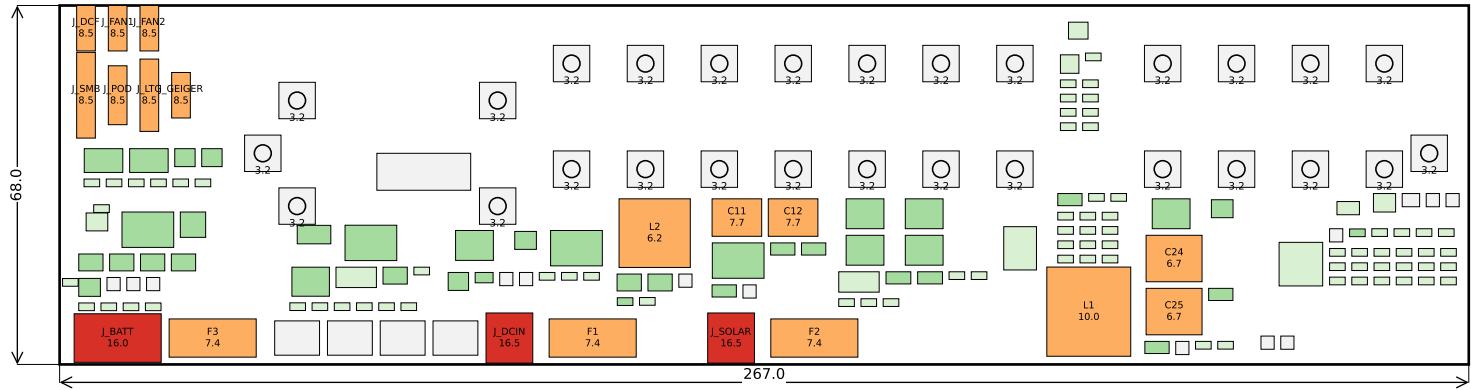
Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
b4fb79d90d2f v2/cad/zstack.json
1ba75de7ceef v2/cad/zstack.py
929bf82d2bf6 v2/ecad/pcb-d-aprs-d9/pcb-d-aprs.kicad_pcb
3fad2fb95f58 v2/cad/zstack-models.json

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 10)
from the files named left; regenerate with v2/cad/README.md.
The editable sources govern; this sheet is a copy.

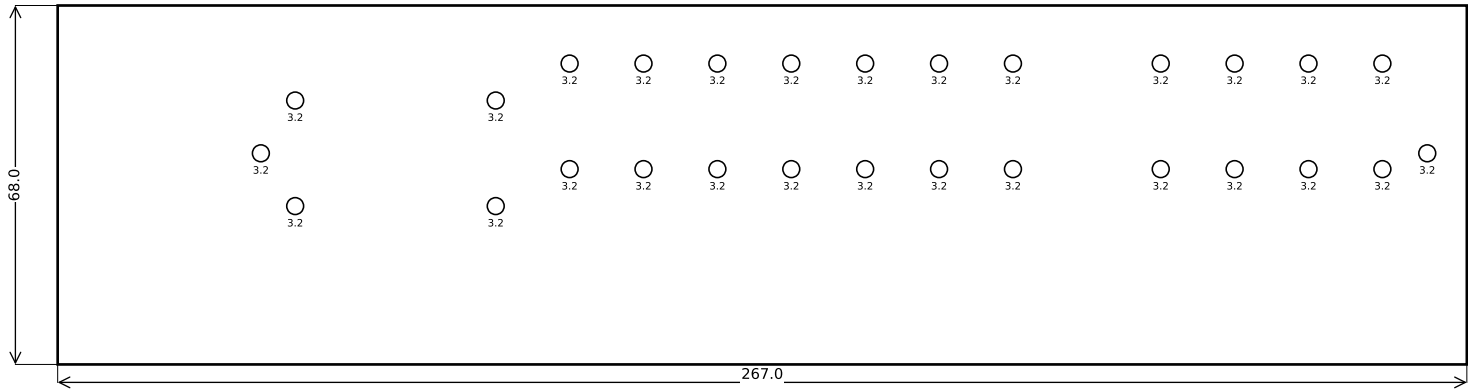
Sheet 10 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

TOP SIDE (seen from above)



BOTTOM SIDE (seen from above, through the board: not mirrored)



v2/ecad/pcb-e1-dock-e7/pcb-e1-dock.kicad_pcb sha256 a462ac2620b9b8d3 (gen_pcb_e.py OX, OY; on the dock strip's VHB pads)
outline X -149.00..118.00, Y -113.00..-45.00 (case frame), thickness 1.60 (the file's), Z 1.10..2.70
tallest top: J.SOLAR 16.50 (JST_VH_B2P-VH 1x02 P3.96mm Vertical); tallest bottom: none
parts 189 by height source: BODYLESS 47, CLASS 2, MAKER 2, MODEL 138
holes: H1 3.2, H5 3.2, H3 3.2, H6 3.2, H4 3.2, H7 3.2, H8 3.2, H9 3.2, H10 3.2, H11 3.2, H12 3.2, H13 3.2, H14 3.2, H15 3.2, H16 3.2, H17 3.2, H18 3.2, H19 3.2, H20 3.2, H21 3.2, H22 3.2, H23 3.2, H24 3.2, H25 3.2, H26 3.2, H27 3.2, H28 3.2, H2 3.2
colour by height above the board surface: no body, <=1, <=3, <=6, <=10, >10 mm (light grey, greens, yellow, orange, red); grey hatched: a height another record owns (COVERED by a module envelope, a PANEL part hanging from the plate, a contact that MATES across a bay); magenta cross-hatched: TBD; B's module envelopes dashed.

HOLES 28 of 28 (drill 2.0 or more), centre in the case frame, mm									
ref	X	Y	drill	what	ref	X	Y	drill	what
H1	-110.50	-73.00	3.20	M3 rod pass-through, PCB-A standoff stands here	H20	32.00	-56.00	3.20	M3, float clamp P2P B
H5	-104.00	-83.00	3.20	M3 standoff, raised block	H21	60.00	-76.00	3.20	M3, float clamp 5G MAIN
H3	-104.00	-63.00	3.20	M3 standoff, raised block	H22	60.00	-56.00	3.20	M3, float clamp 5G MAIN
H6	-66.00	-83.00	3.20	M3 standoff, raised block	H23	74.00	-76.00	3.20	M3, float clamp 5G DIV
H4	-66.00	-63.00	3.20	M3 standoff, raised block	H24	74.00	-56.00	3.20	M3, float clamp 5G DIV
H7	-52.00	-76.00	3.20	M3, float clamp VHF	H25	88.00	-76.00	3.20	M3, float clamp IRIDIUM
H8	-52.00	-56.00	3.20	M3, float clamp VHF	H26	88.00	-56.00	3.20	M3, float clamp IRIDIUM
H9	-38.00	-76.00	3.20	M3, float clamp HF	H27	102.00	-76.00	3.20	M3, float clamp LORA
H10	-38.00	-56.00	3.20	M3, float clamp HF	H28	102.00	-56.00	3.20	M3, float clamp LORA
H11	-24.00	-76.00	3.20	M3, float clamp WIFI 2.4	H2	110.50	-73.00	3.20	M3 rod pass-through, PCB-A standoff stands here
H12	-24.00	-56.00	3.20	M3, float clamp WIFI 2.4					
H13	-10.00	-76.00	3.20	M3, float clamp GNSS					
H14	-10.00	-56.00	3.20	M3, float clamp GNSS					
H15	4.00	-76.00	3.20	M3, float clamp SDR					
H16	4.00	-56.00	3.20	M3, float clamp SDR					
H17	18.00	-76.00	3.20	M3, float clamp P2P A					
H18	18.00	-56.00	3.20	M3, float clamp P2P A					
H19	32.00	-76.00	3.20	M3, float clamp P2P B					

TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines ± 0.10 ; the 3.0 face plate ± 0.13 and the 6.0 entry plates ± 0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height ± 0.10 ; rebate depth and line ± 0.10 ; spot-face floors ± 0.10 .
Peli: the 1450PF sheet 1453-314-000 rev A states ± 0.76 on one-decimal and ± 0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
Heights are nominal library bodies at the committed placements; a board's plan tolerance is JLC's outline class (± 0.2 , not held) and the stack's placement ± 1.0 by hand (INFERRED).

MeshSat field kit V2, Peli 1450 case set

Board envelope E dock strip: outline, holes and parts by height on both sides, read from the committed board

MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

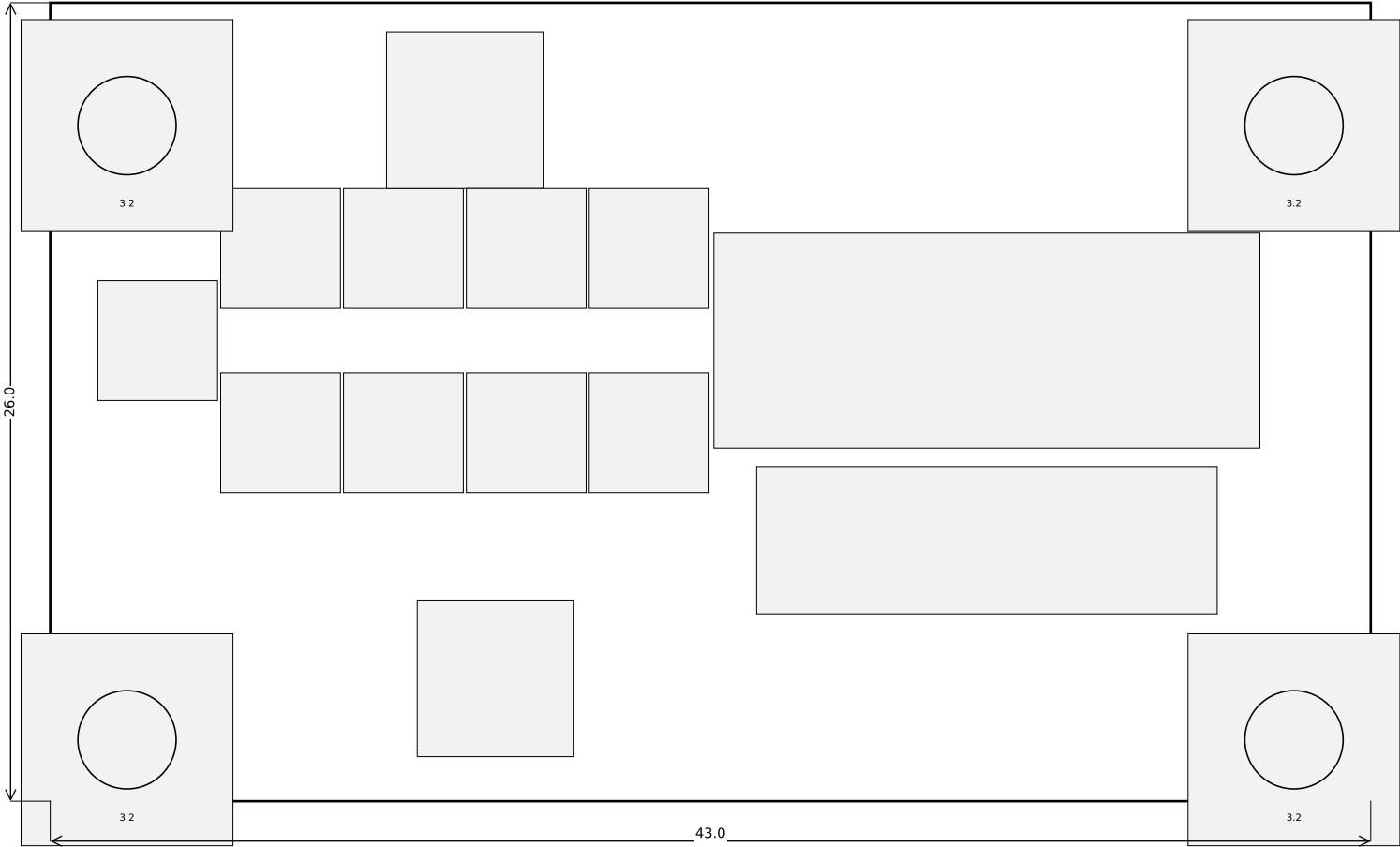
Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
b4fb79d90d2f v2/cad/zstack.json
1ba75de7ceef v2/cad/zstack.py
a462ac2620b9 v2/ecad/pcb-e1-dock-e7/pcb-e1-dock.kicad_pcb
3fad2fb95f58 v2/cad/zstack-models.json

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 11)
from the files named left; regenerate with v2/cad/README.md.
The editable sources govern; this sheet is a copy.

Sheet 11 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

TOP SIDE (seen from above)



BOTTOM SIDE (seen from above, through the board: not mirrored)



v2/ecad/pcb-e5-block/pcb-e5-block.kicad_pcb sha256 686b29a734c55b9a (gen_pcb_e5.py: face 7.4 above the dock strip on M3 standoffs (the docstring's figure; a 6 mm standoff and 1.6 board give 7.6, INFERRED))
outline X -106.50...-63.50, Y -85.00...-59.00 (case frame), thickness 1.60 (the file's), Z 8.50...10.10
tallest top: H4 0.00 (MountingHole 3.2mm_M3); tallest bottom: none
parts 17 by height source: BODYLESS 17
holes: H3 3.2, H1 3.2, H4 3.2, H2 3.2
colour by height above the board surface: no body, <=1, <=3, <=6, <=10, >10 mm (light grey, greens, yellow, orange, red); grey hatched: a height another record owns (COVERED by a module envelope, a PANEL part hanging from the plate, a contact that MATES across a bay); magenta cross-hatched: TBD; B's module envelopes dashed.

HOLES 4 of 4 (drill 2.0 or more), centre in the case frame, mm				
ref	X	Y	drill	what
H3	-104.00	-83.00	3.20	M3 standoff 6 mm to the dock strip
H1	-104.00	-63.00	3.20	M3 standoff 6 mm to the dock strip
H4	-66.00	-83.00	3.20	M3 standoff 6 mm to the dock strip
H2	-66.00	-63.00	3.20	M3 standoff 6 mm to the dock strip

TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines ± 0.10 ; the 3.0 face plate ± 0.13 and the 6.0 entry plates ± 0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height ± 0.10 ; rebate depth and line ± 0.10 ; spot-face floors ± 0.10 .
Peli: the 1450PF sheet 1453-314-000 rev A states ± 0.76 on one-decimal and ± 0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
Heights are nominal library bodies at the committed placements; a board's plan tolerance is JLC's outline class (± 0.2 , not held) and the stack's placement ± 1.0 by hand (INFERRED).

MeshSat field kit V2, Peli 1450 case set

Board envelope E5 dock block: outline, holes and parts by height on both sides, read from the committed board

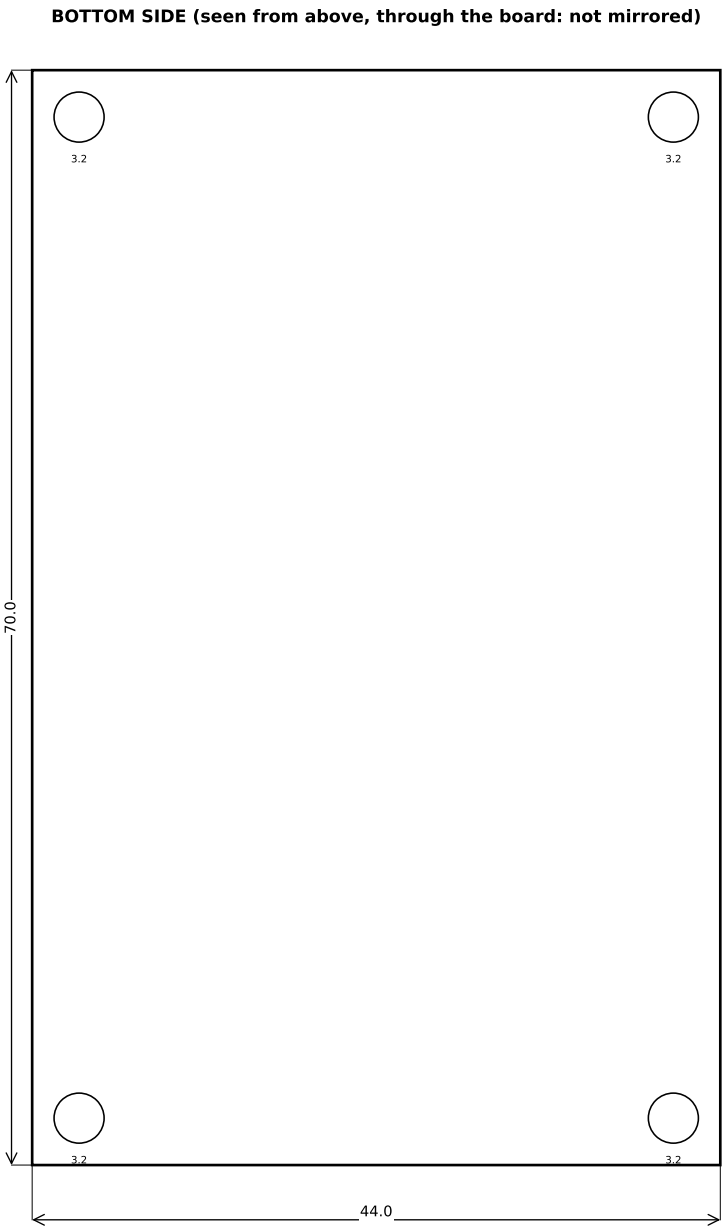
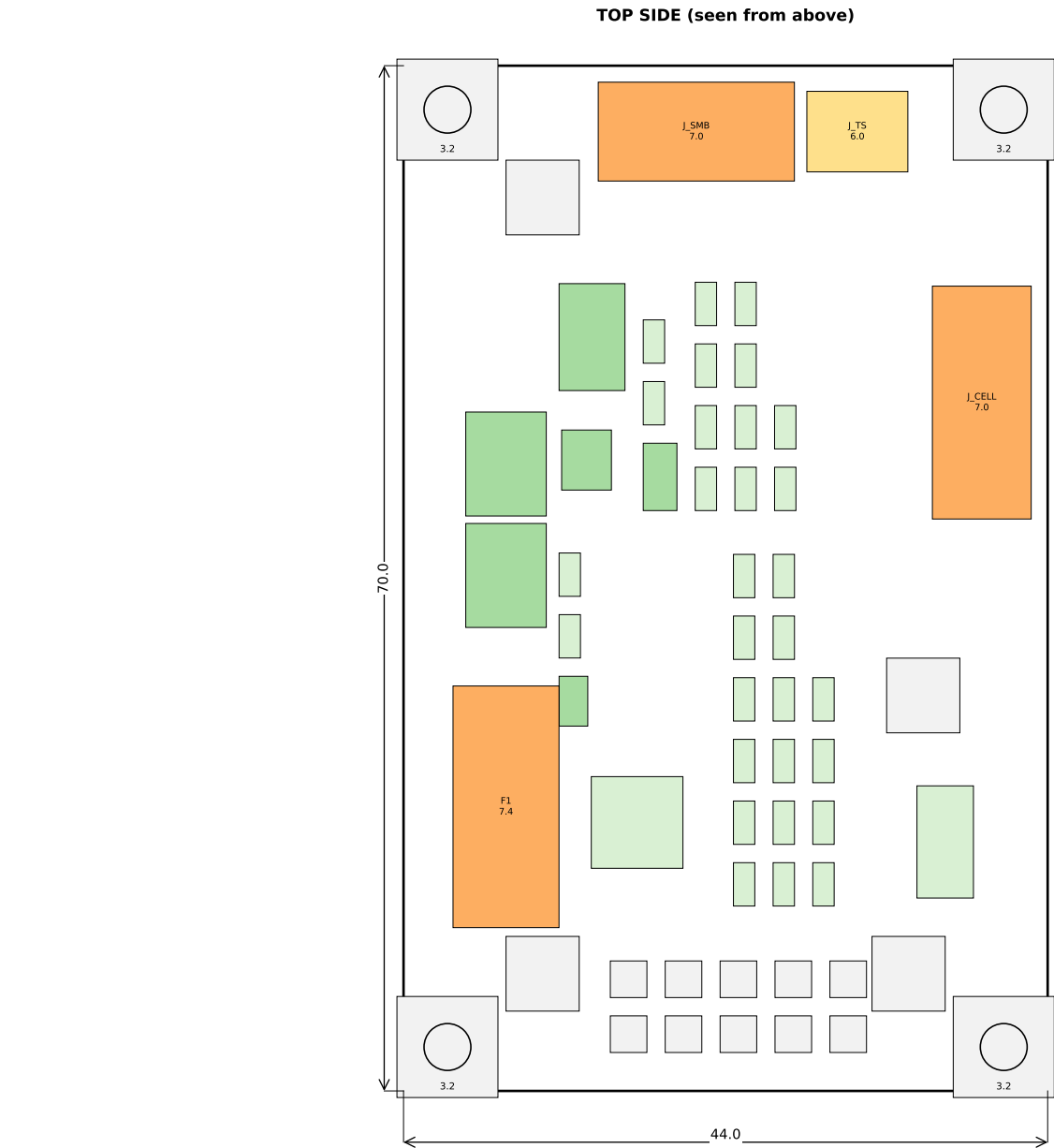
MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
b4fb79d90d2f v2/cad/zstack.json
1ba75de7ceef v2/cad/zstack.py
686b29a734c5 v2/ecad/pcb-e5-block/pcb-e5-block.kicad_pcb
3fad2fb95f58 v2/cad/zstack-models.json

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 12)
from the files named left; regenerate with v2/cad/README.md.
The editable sources govern; this sheet is a copy.

Sheet 12 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking



v2/ecad/pcb-p-pack-p2/pcb-p-pack.kicad_pcb sha256 d79865e7b1aceb95 (INFERRED: the south end of the pack group centred in Y (CASE-MARGINS.md M5), its long side along the pack; Z not set until the hold-down is designed)
outline X 128.32..172.32, Y -102.75...-32.75 (case frame), thickness 1.60 (the file's), Z not set (pack hold-down undesigned)
tallest top: F1 7.37 (Fuseholder Blade Mini Keystone 3568); tallest bottom: none
parts 60 by height source: BODYLESS 18, CLASS 2, MODEL 40
holes: H3 3.2, H4 3.2, H1 3.2, H2 3.2
colour by height above the board surface: no body, <=1, <=3, <=6, <=10, >10 mm (light grey, greens, yellow, orange, red); grey hatched: a height another record owns (COVERED by a module envelope, a PANEL part hanging from the plate, a contact that MATES across a bay); magenta cross-hatched: TBD; B's module envelopes dashed.

HOLES 4 of 4 (drill 2.0 or more), centre in the case frame, mm					
ref	X	Y	drill	what	
H3	131.32	-99.75	3.20	M3 into the enclosure boss	
H4	131.32	-35.75	3.20	M3 into the enclosure boss	
H1	169.32	-99.75	3.20	M3 into the enclosure boss	
H2	169.32	-35.75	3.20	M3 into the enclosure boss	

TOLERANCES AND STATUS
Dimensions in mm. Kit parts (this sheet): machined positions and outlines +-0.10; the 3.0 face plate +-0.13 and the 6.0 entry plates +-0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +-0.10; rebate depth and line +-0.10; spot-face floors +-0.10.
Peli: the 1450PF sheet 1453-314-000 rev A states +-0.76 on one-decimal and +-0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).
Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.
Heights are nominal library bodies at the committed placements; a board's plan tolerance is JLC's outline class (+-0.2, not held) and the stack's placement +-1.0 by hand (INFERRED).

MeshSat field kit V2, Peli 1450 case set

Board envelope P pack BMS: outline, holes and parts by height on both sides, read from the committed board

MESH SAT-1357. PROTOTYPE DESIGN: nothing made, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026 (SC-07).

Source: tree a8652172 plus the case release's changes (fnd/hc7); inputs by sha256 (first 12):
b4fb79d90d2f v2/cad/zstack.json
1ba75de7ceef v2/cad/zstack.py
d79865e7b1ac v2/ecad/pcb-p-pack-p2/pcb-p-pack.kicad_pcb
3fad2fb95f58 v2/cad/zstack-models.json

Generated 2026-09-27 by v2/cad/case_drawings.py (sheet 13)
from the files named left; regenerate with v2/cad/README.md.
The editable sources govern; this sheet is a copy.

Sheet 13 of 14

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking

Technical drawing of a pocket unit. The drawing shows a yellow rectangular unit with a dashed border. The unit is labeled "QMX unit 95 x 63 x 25 (generator's figure, not in the held manual: TBD)". The unit is mounted on a wall. The wall is labeled "lid's inner ceiling Z 154.44 (rim 108.97 + lid 45.47, Peli STEP)". The unit's height is labeled "28.0". The unit's width is labeled "69.0". The unit's depth is labeled "open face of the pocket Z 126.44". The unit's top edge is labeled "M19 + 2.08". The unit's bottom edge is labeled "SW_PI: cap height above the face TBD (C&K sheet), M3". The unit's left edge is labeled "M3 + 19.92". The unit's right edge is labeled "face top Z 106.52 (104.77..108.27, M20)".

Technical drawing of a rectangular plate with the following specifications:

- Overall width: 123.0 over the tabs (body 101.0)
- Overall height: 69.0
- Notches (red) on the left side.
- Slots: 3 x 18 at 30.0.
- Pocket: 97 x 65 R3 (dashed), outer R4; tabs 12 x 8 x 3.
- Bottom width: 4 x $\phi 3.4$ on 116.0 x 61.0.

Technical drawing of a rectangular box. The drawing shows a top view and a side view. The top view is a rectangle with a width of 81.0 and a height of 28.0. The side view is a rectangle with a width of 2.6. The drawing is labeled with dimensions and text: "side walls open above the lip, 81.0" (top), "28.0" (left), "2.6" (right), and "SIDE (along the length); tabs 3 thick on the tab face (z 0)" (bottom).

Diagram of a notched end of a beam. The total length is 69.0. A notch of width 12 is located at a distance of 36 from the left end. The notch depth is 0.0.

- Margin rows this sheet's features enter (frame_seal.out): key, verdict, min, nominal, worst, worst with unstated x2
- | | | | | | | |
|-----|------|---|--------|--------|--------|--|
| M3 | OPEN | 1.00 | +19.92 | +16.39 | +14.11 | space under the QMX tray for the face parts (lid STEP 45.47; web 44.45 in the worst) |
| M19 | MET | 1.00 | +2.08 | +1.70 | +1.32 | QMX tray east edge inside the lid's flat ceiling (C5) |
| M20 | OPEN | the frame's seat: bottom Z 86.00 nominal, 84.38..87.62 on the legs (T2) | | | | |

Dimensions in mm. Kit parts (this sheet): machined positions and outlines +0.10; the 3.0 face plate +0.13 and the 6.0 entry plates +0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +0.10; rebate depth and line +0.30; spot-face floors +0.10.

Pel: the 1450PF sheet 1453-314-000 rev A states +0.76 on one-decimal and +0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).

Design basis only (v2/docs/CASE-MARGINS.md section 1. Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity analysis of the 1450PF sheet 1453-314-000 rev A states no tolerance on some of the dimensions (INFERRED, UNSTATED).

Final CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.

This sheet's part is printed; its tolerances are the printer's and are not stated (TBD); the place is marked on the lid by hand from the case centre and the flat ceiling's edge (+1.0, INFERRED). M19 carries the lid's wall allowance.

A3, mm
not to scale for measurement:
dimensions govern; use the
1:1 templates for marking